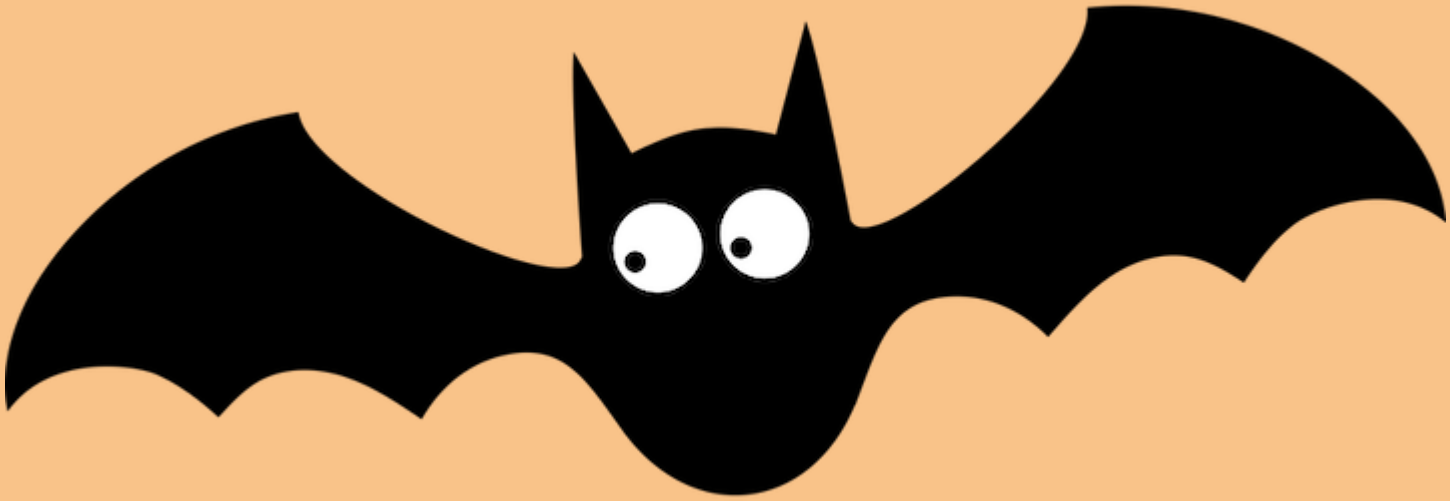


Creating Useful Fake Data

Son Nguyen

Classification Problem

Classification Problem

Classification Problem

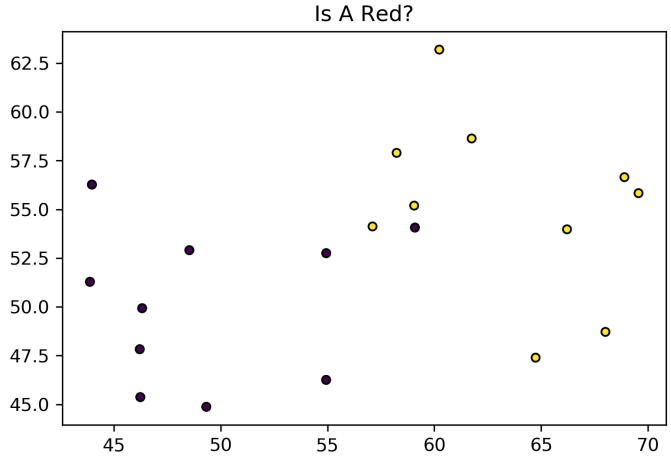
- Credit Card Default?

Classification Problem

- Credit Card Default?
- Or not?

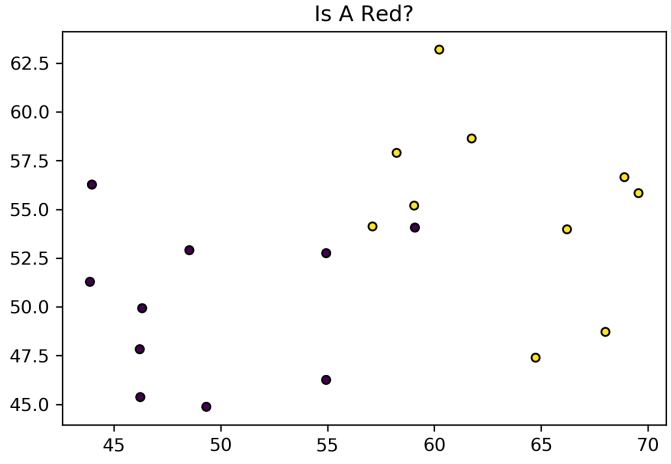
Classification Problem

- Is A yellow?



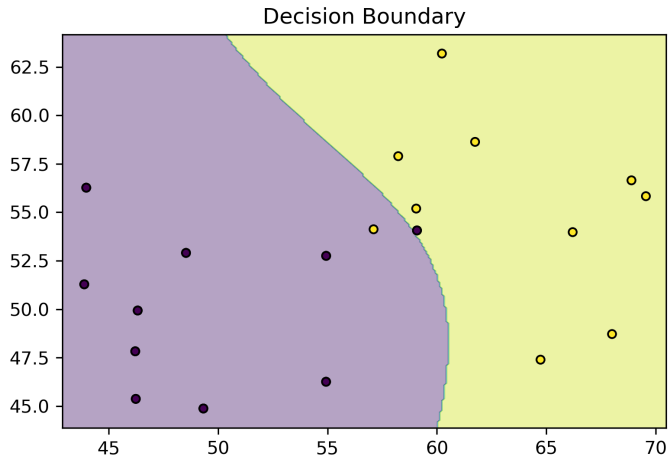
Classification Problem

- Is A yellow?
- Or purple?



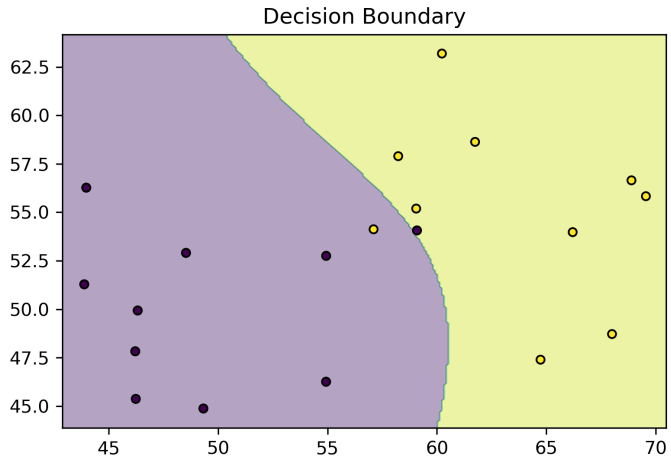
Classification Problem

- Decision Boundary



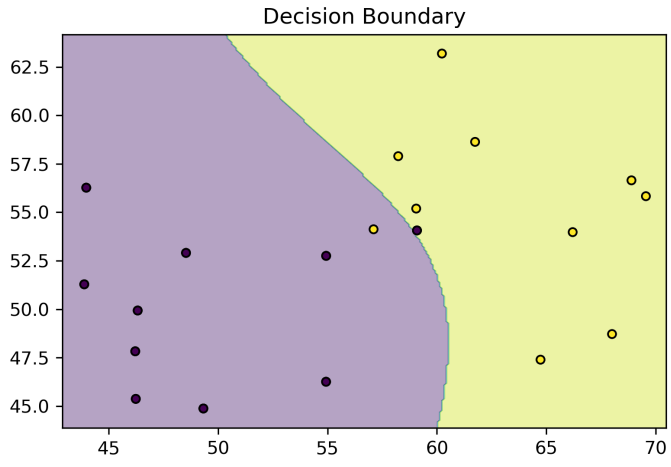
Classification Problem

- Decision Boundary
- of a classification model



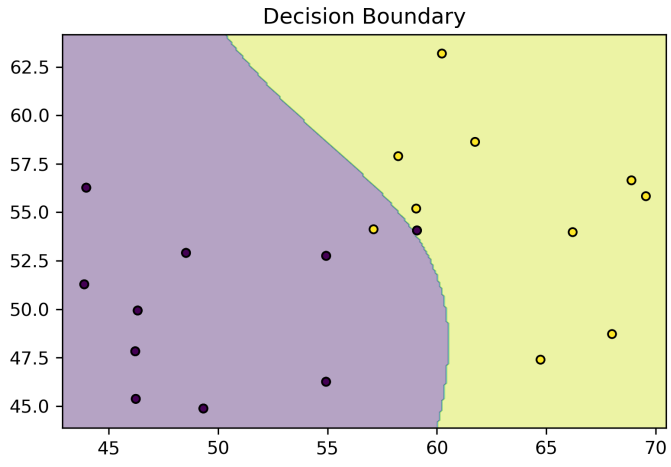
Classification Problem

- Decision Boundary
- of a classification model
- A is...

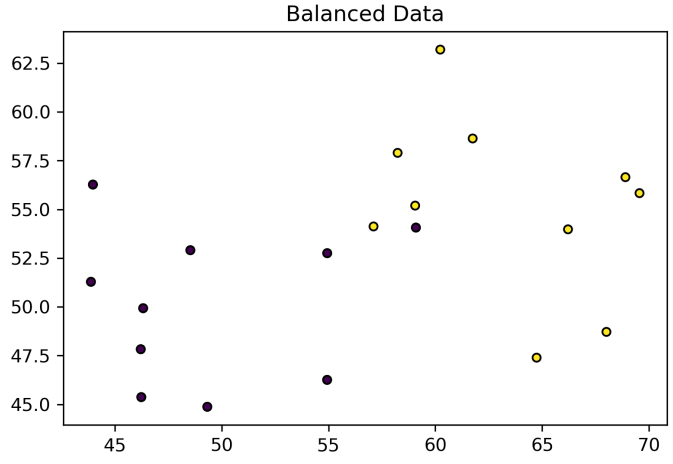


Accuracy

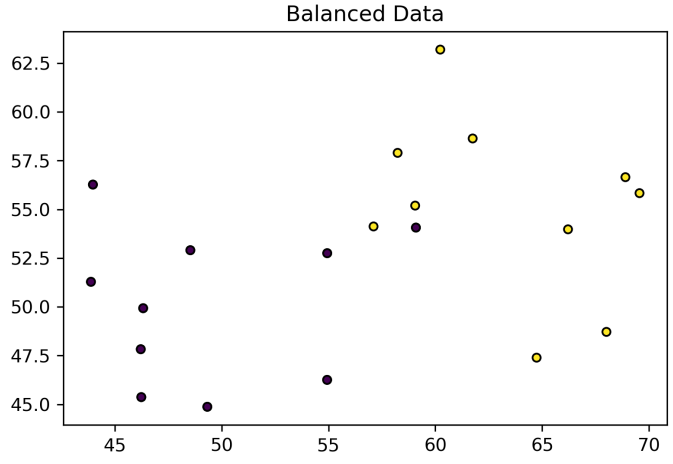
Purple Accuracy	Yellow Accuracy	Balanced Accuracy
0.9	0.9	0.9

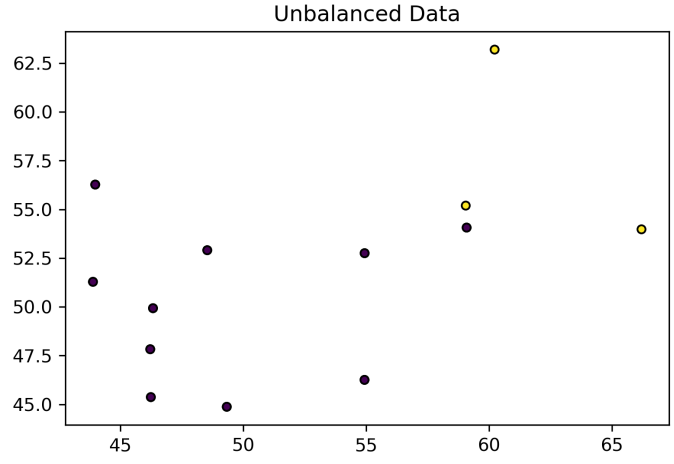


Balance vs. Unbalanced Data

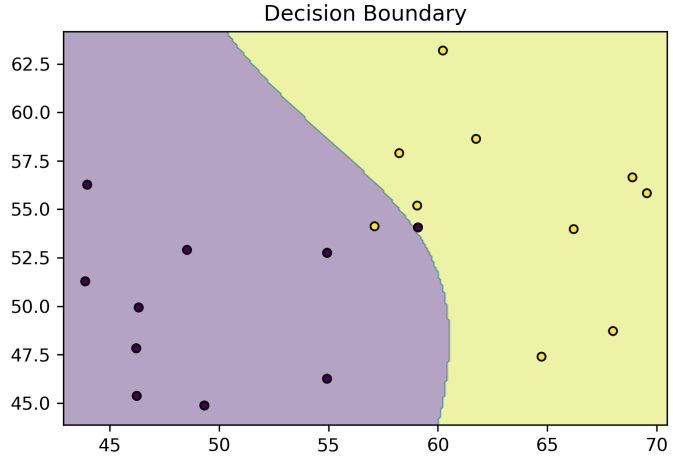


Balance vs. Unbalanced Data

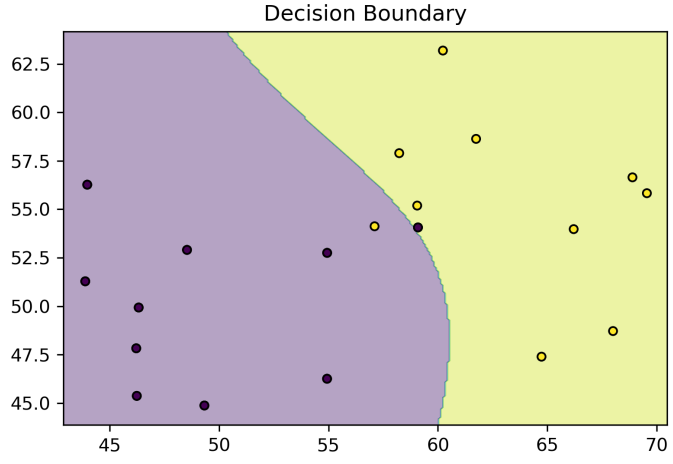


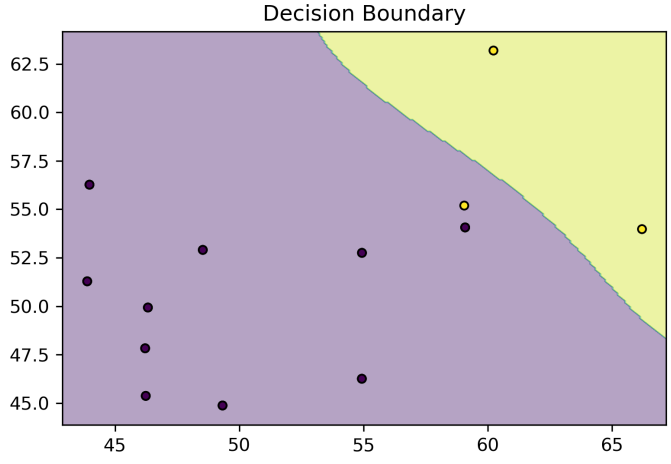


Decision Boundary of Unbalanced Data

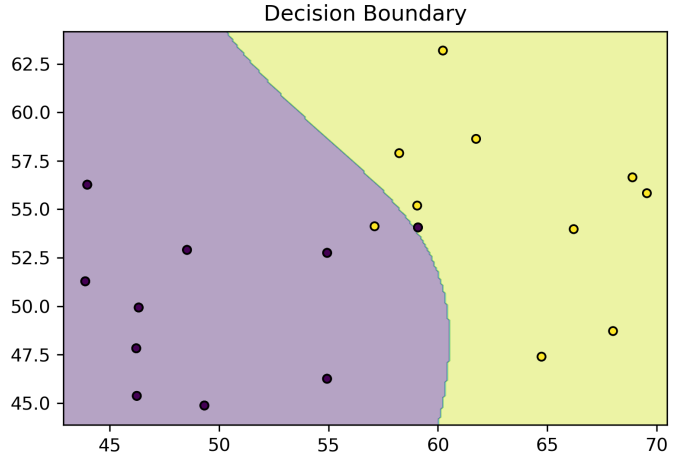


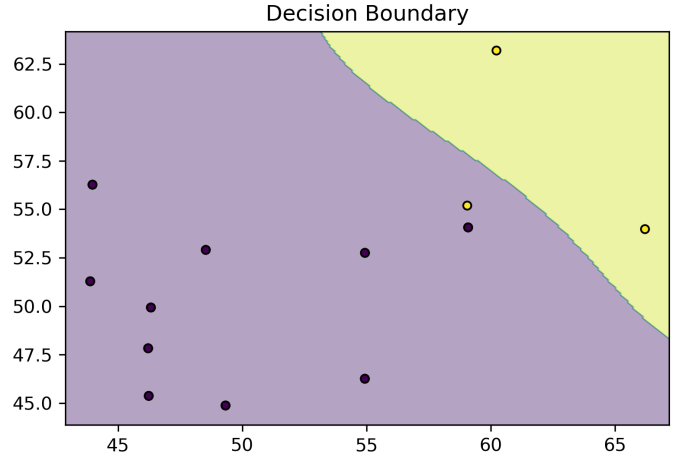
Decision Boundary of Unbalanced Data

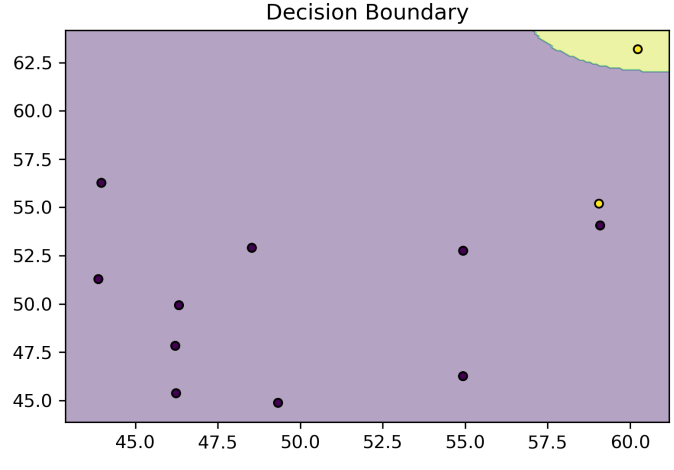




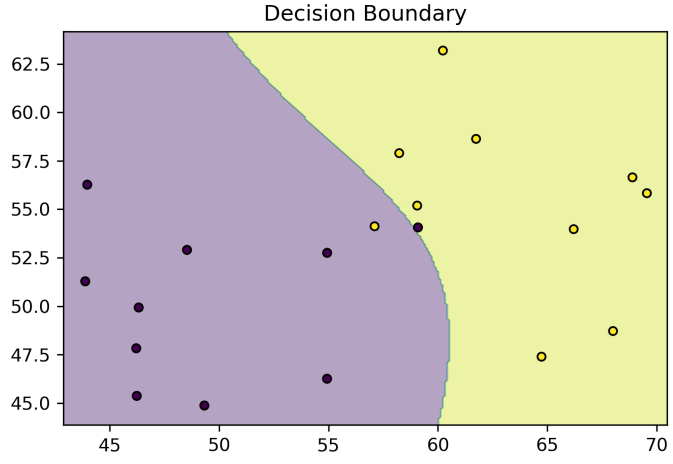
Decision Boundary of Unbalanced Data

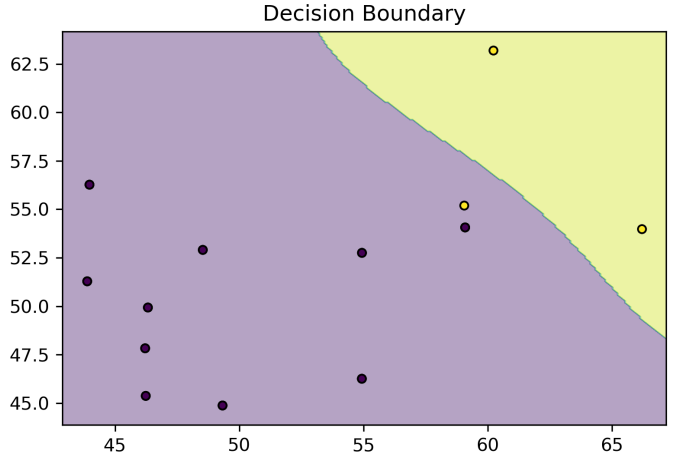


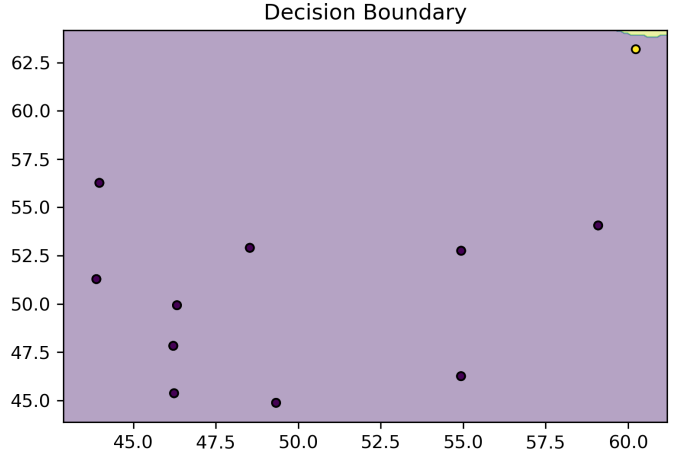


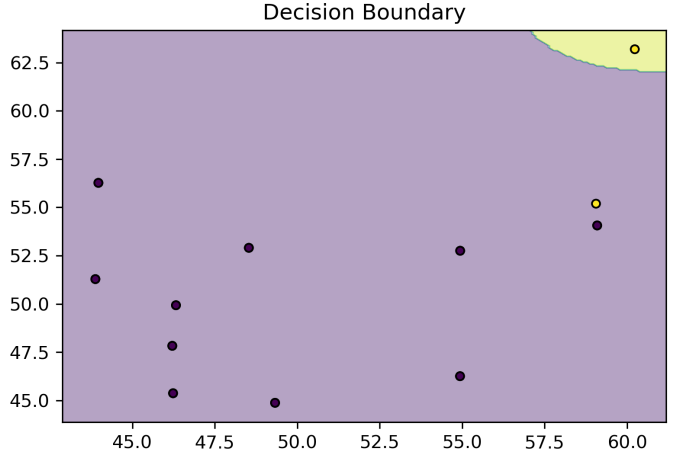


Decision Boundary of Unbalanced Data



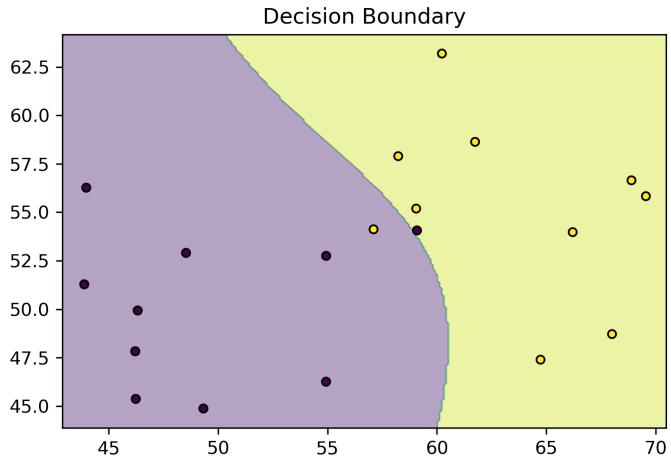






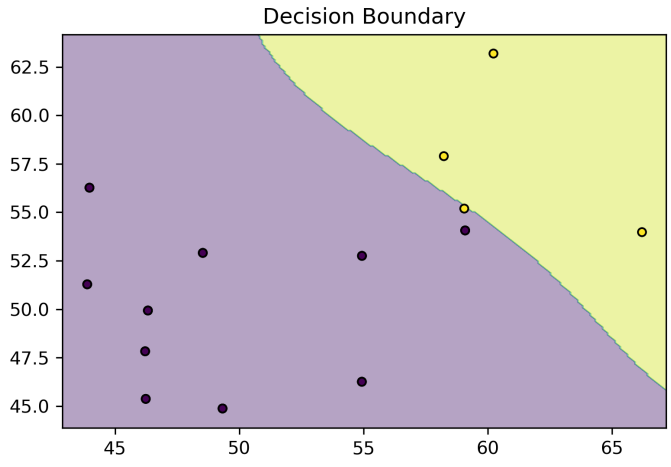
Accuracy

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.9	0.9	0.9



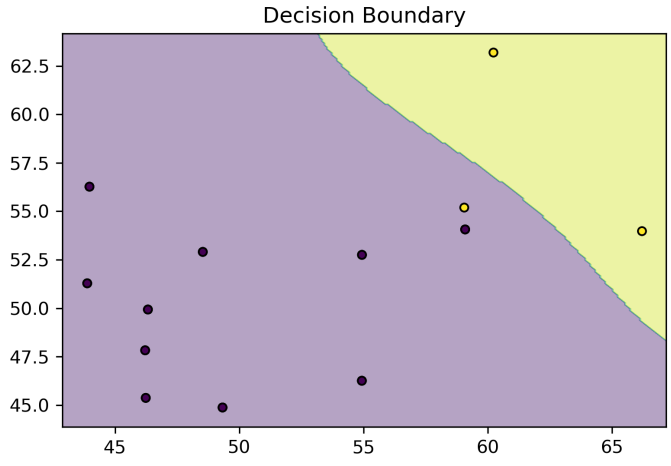
Accuracy

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.75	1	0.875



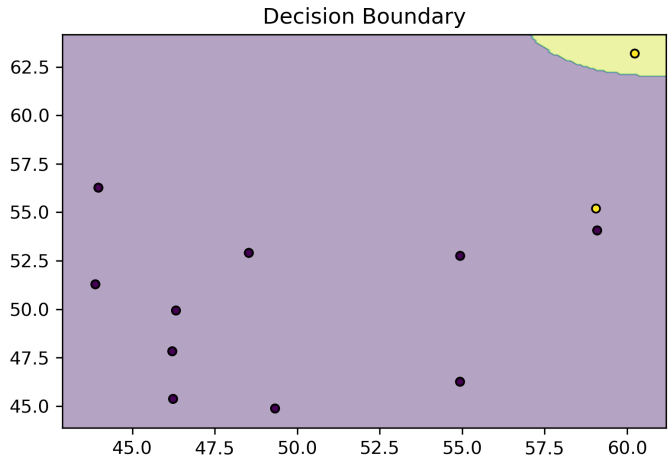
Accuracy

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.666667	1	0.833333



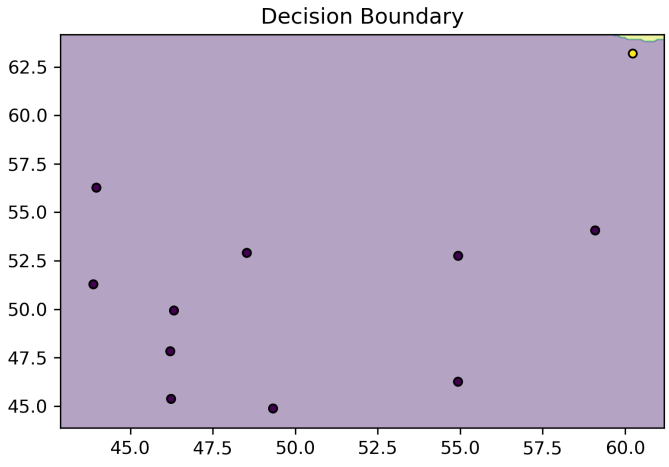
Accuracy

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.5	1	0.75



Accuracy

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0	1	0.5



- It's more difficult to detect yellow points when the imbalance increases
- Yellow points could present
    - Cancer Patients,
    - Credit Card Fraud,
    - Bankrupted Companies,
    - Terrorist
- They are all rare events.

Question

- How to detect the yellow points when there are too few of them?

Strategy 1

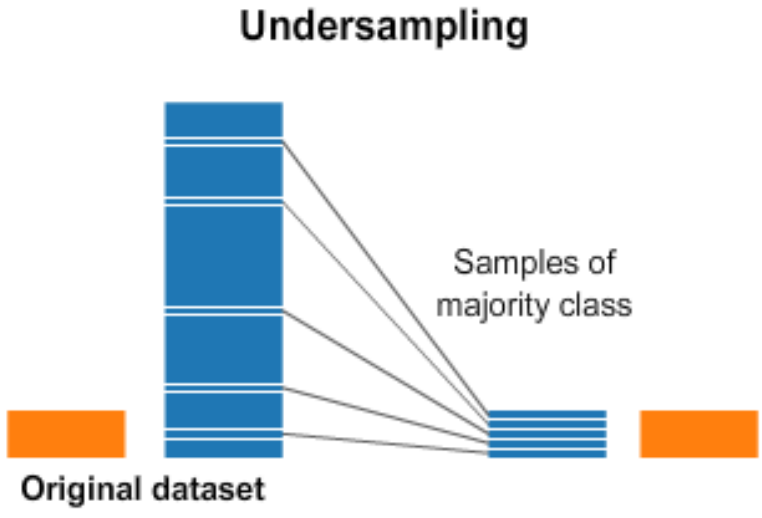
- Remove the purple points so that the data more balanced

Strategy 1

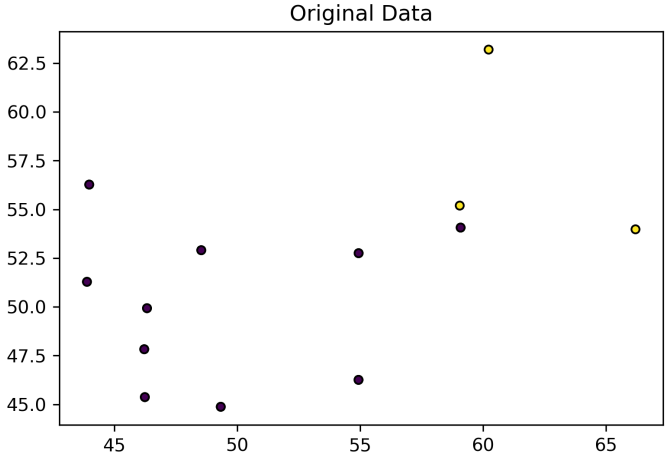
- Remove the purple points so that the data more balanced
- This is called under-sampling

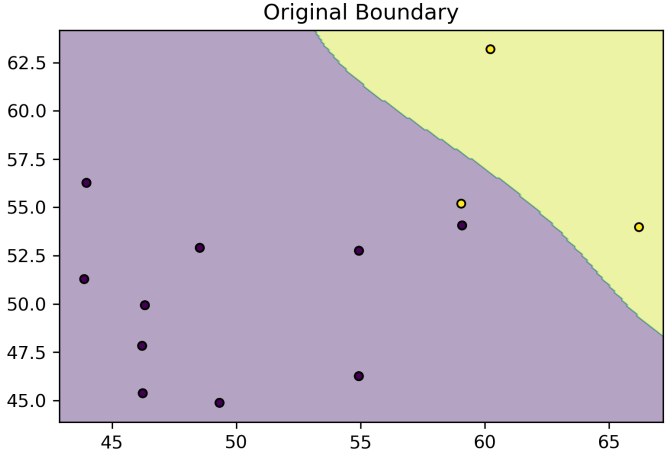
Strategy 1

- Remove the purple points so that the data more balanced
- This is called under-sampling

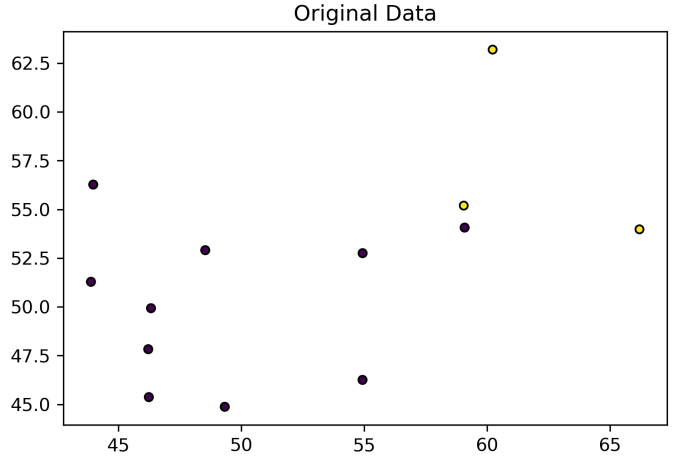


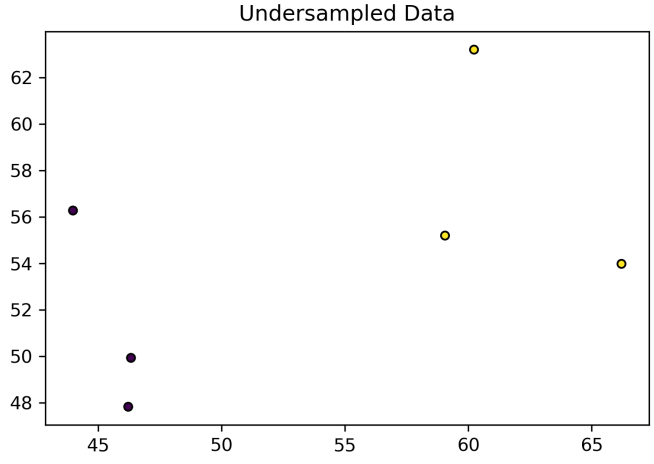
Undersampling

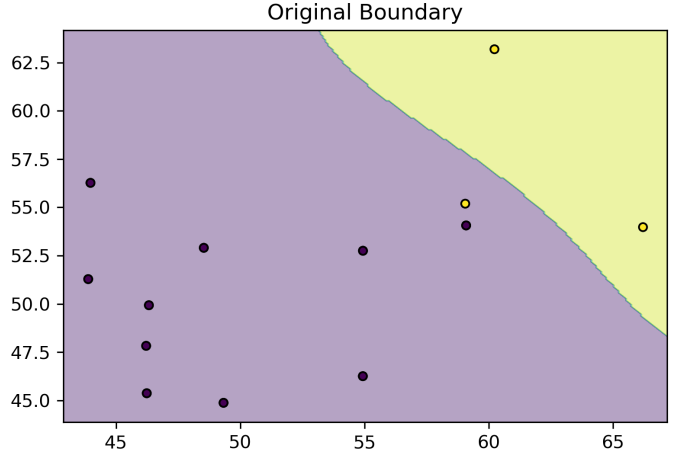




Undersampling







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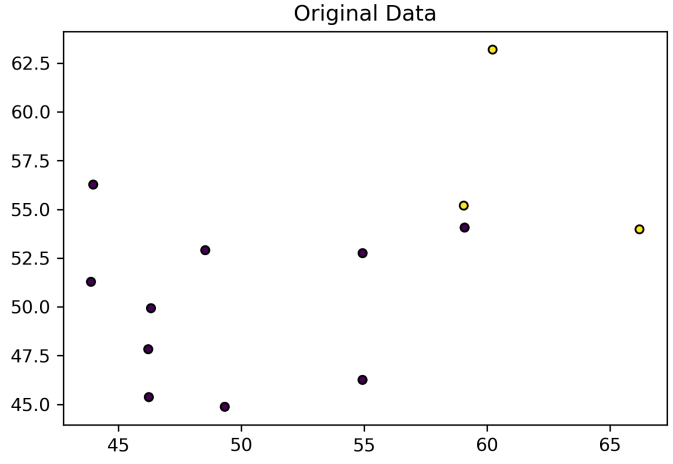
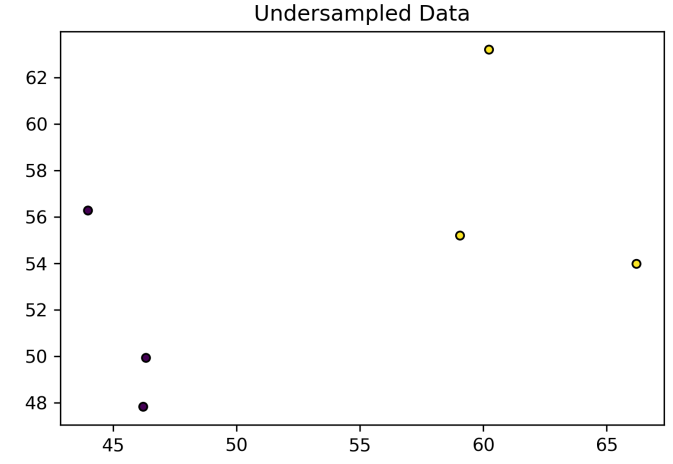
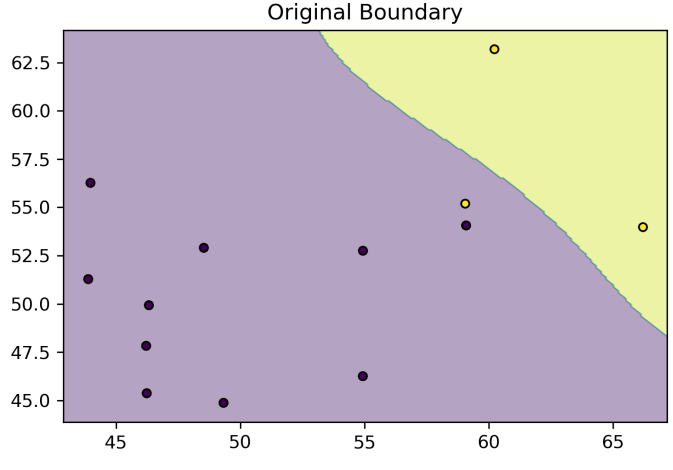
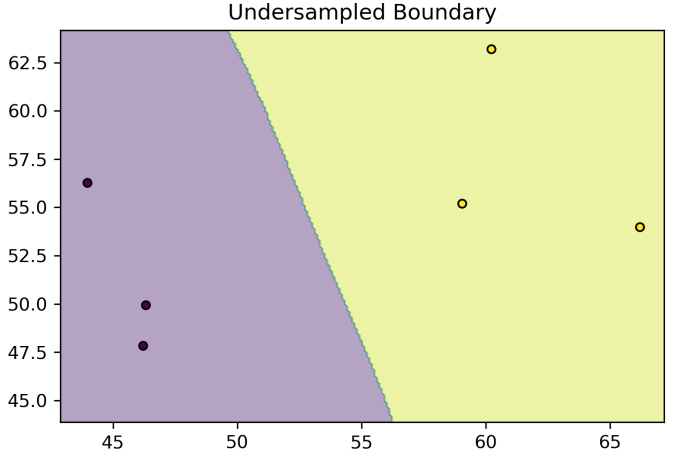
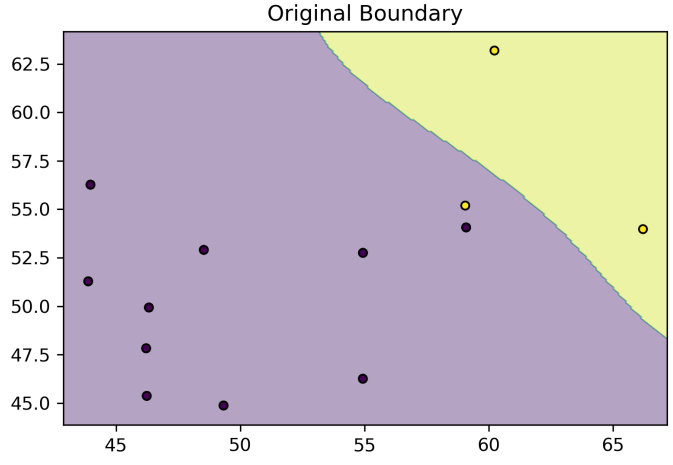


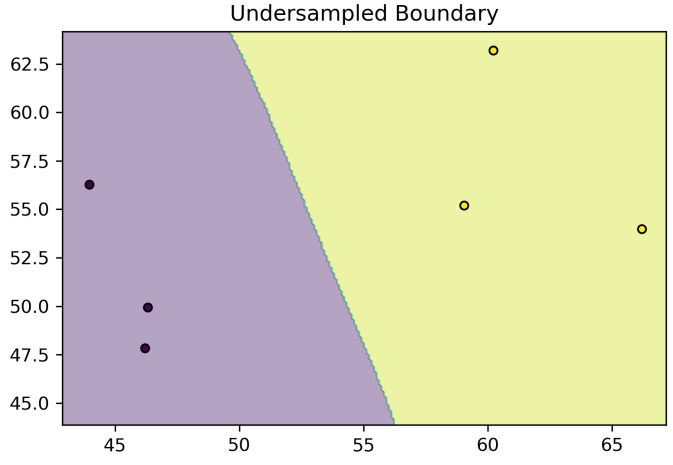






Undersampling

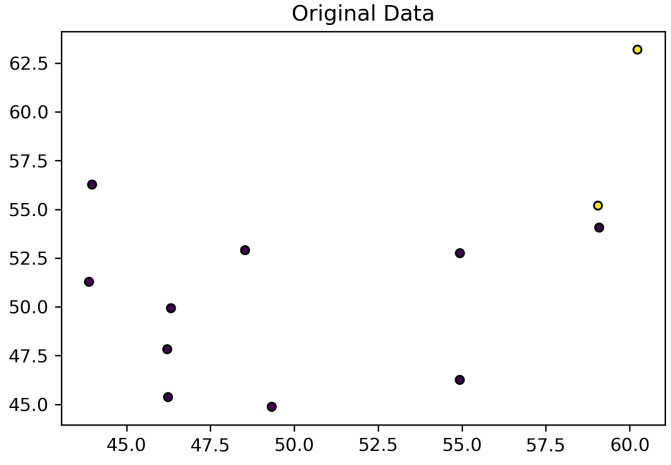


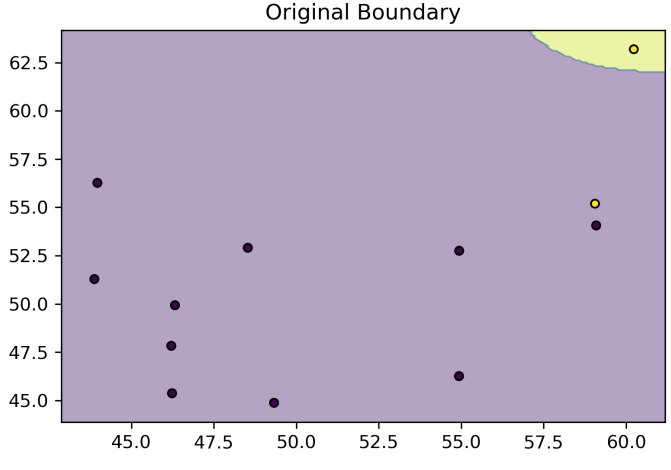


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.666667	1	0.833333

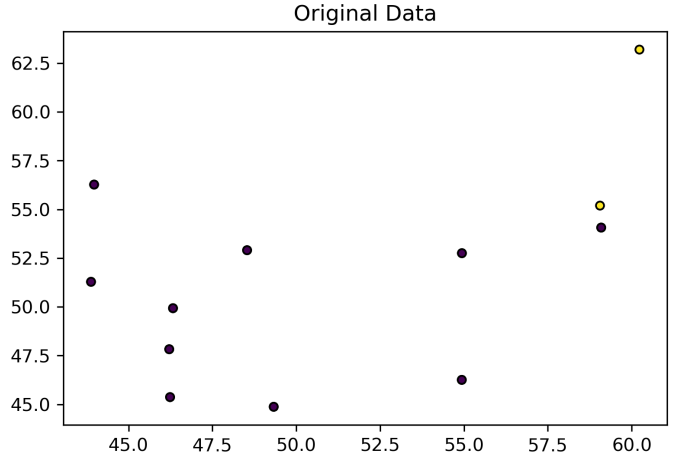
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	0.8	0.9

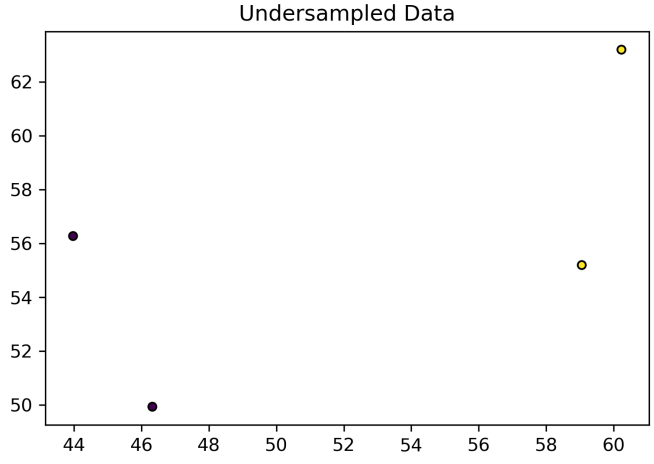
Undersampling

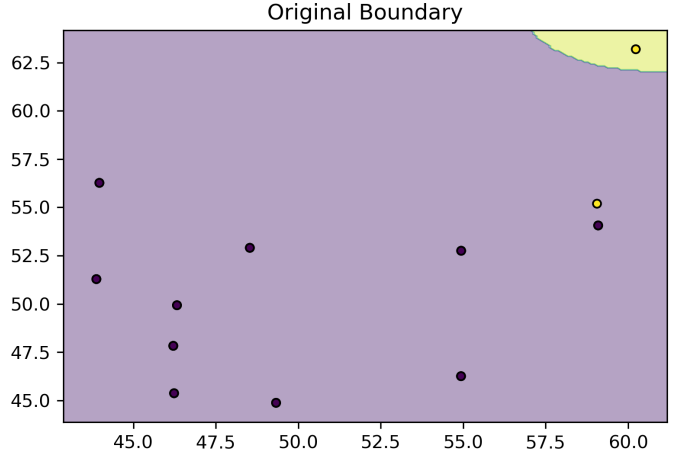




Undersampling







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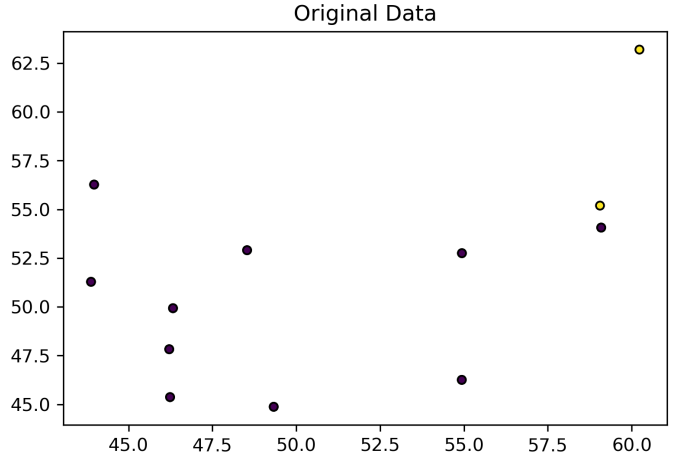
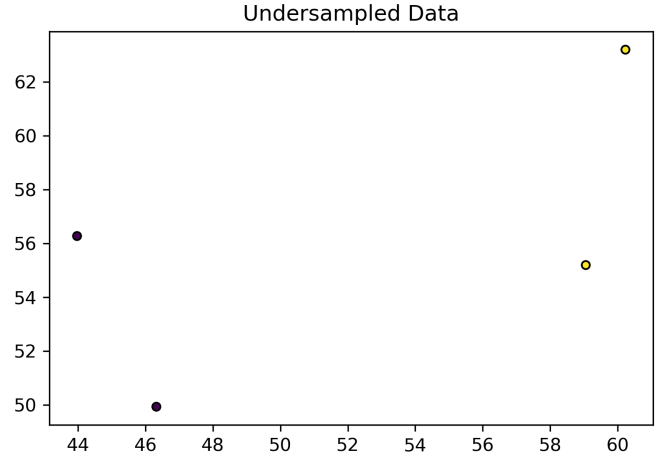
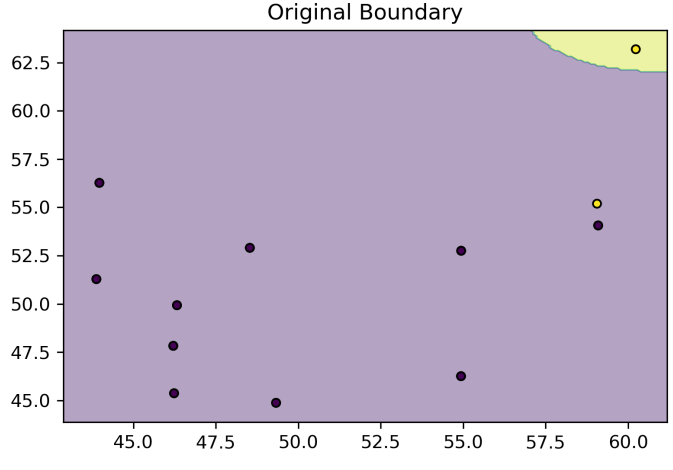
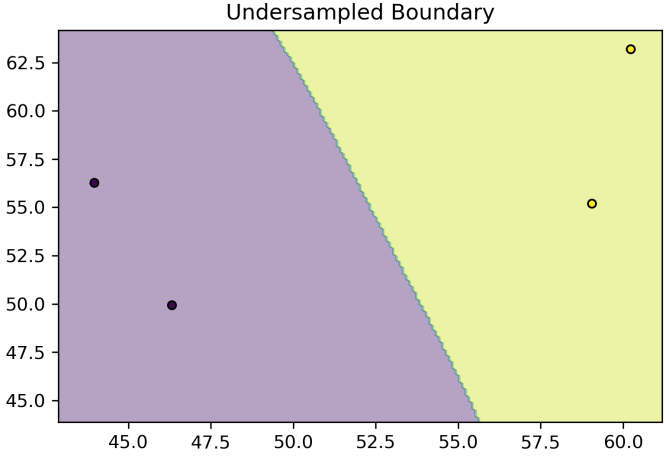
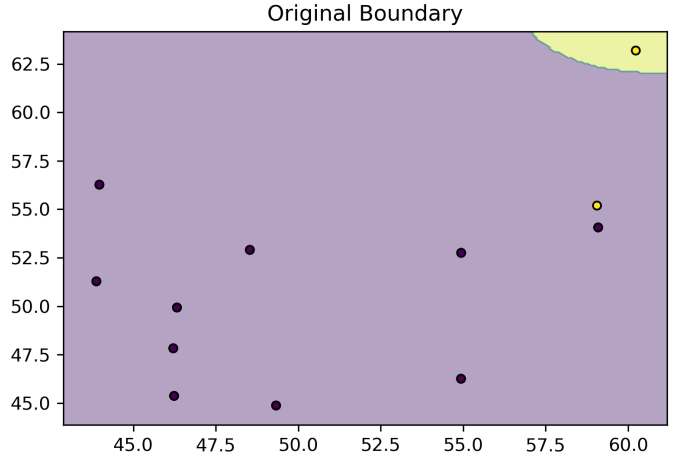


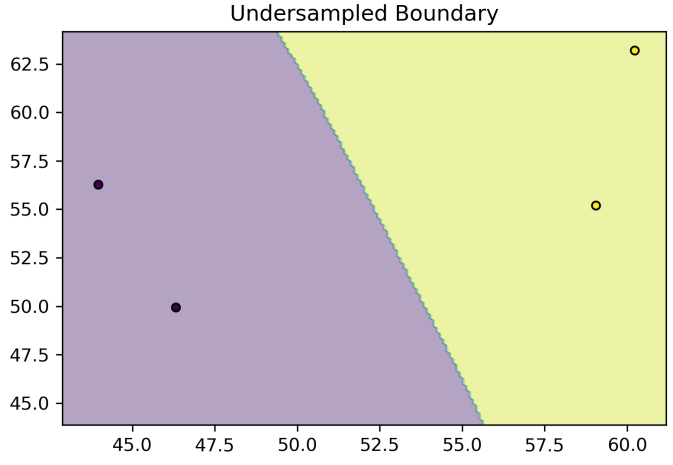






Undersampling

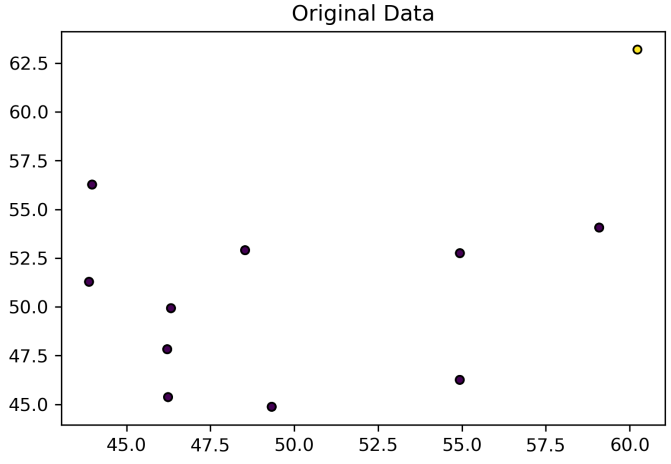


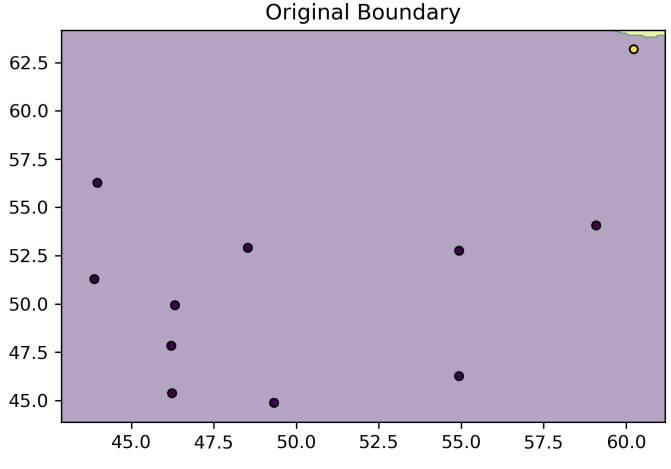


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.5	1	0.75

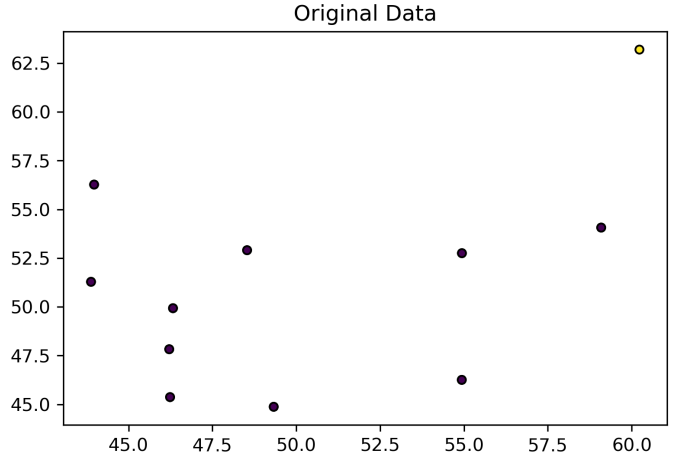
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	0.8	0.9

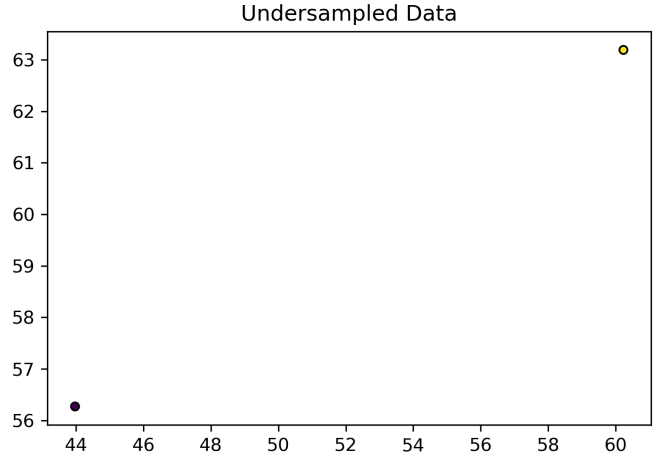
Undersampling

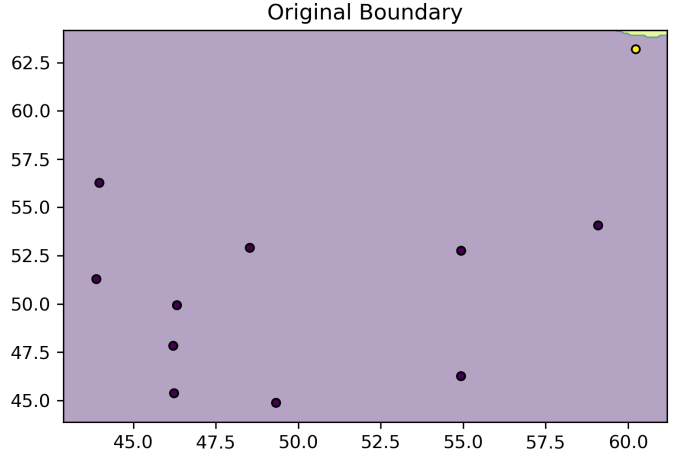




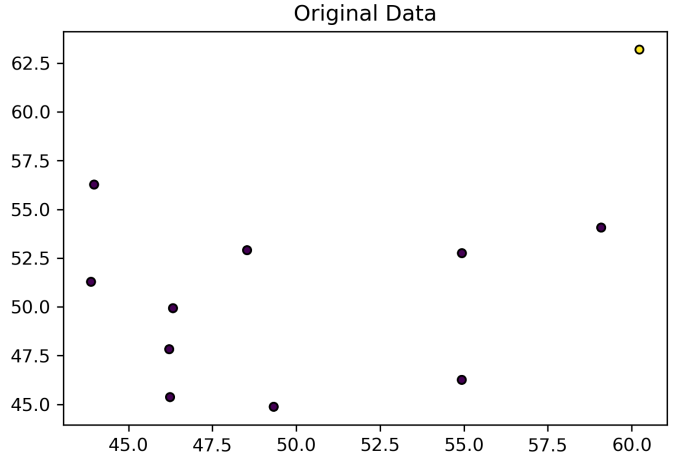
Undersampling

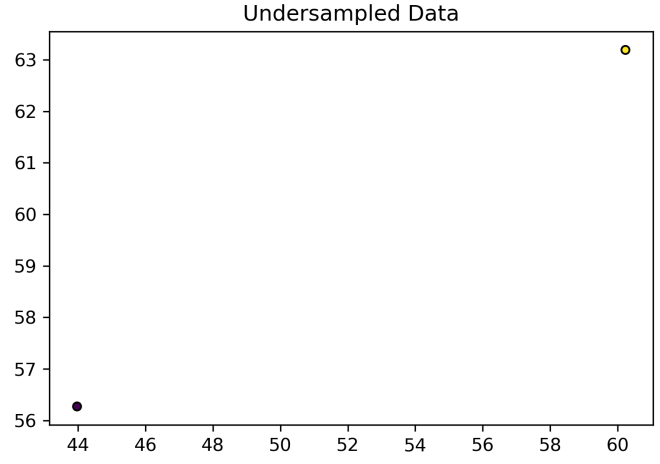


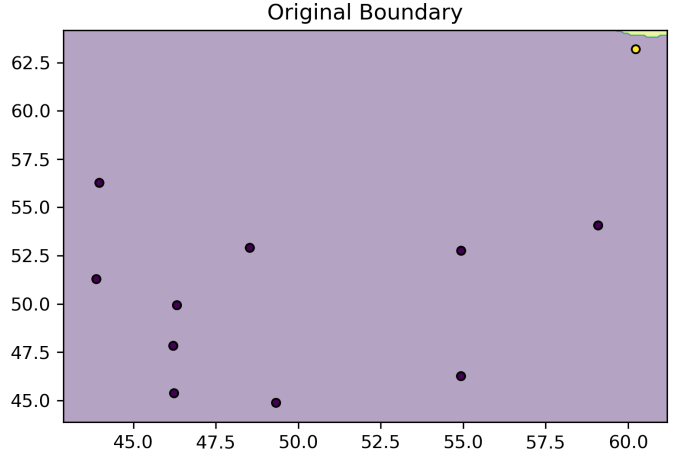


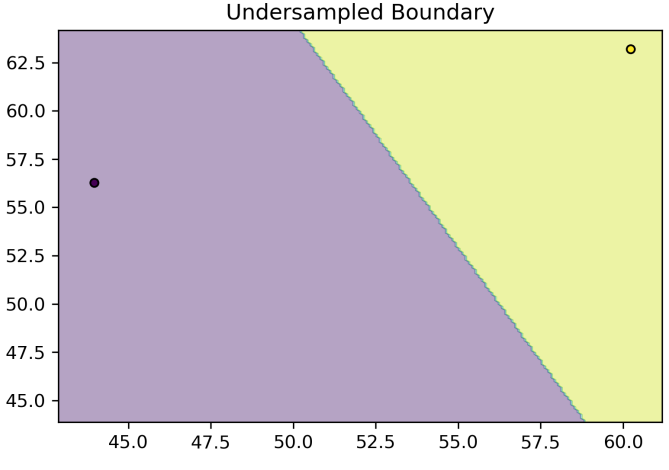


Undersampling

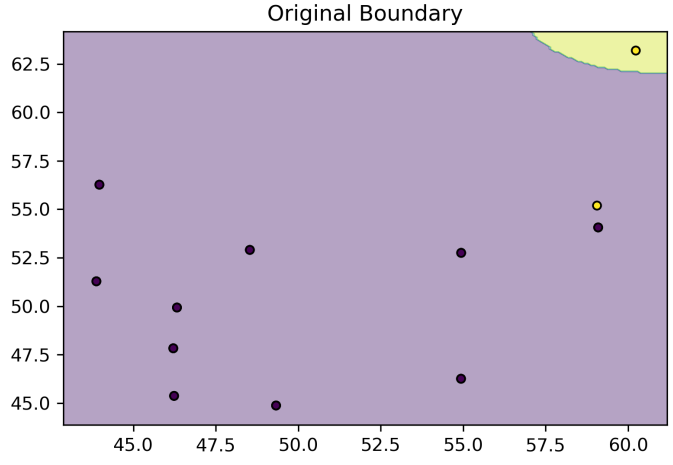


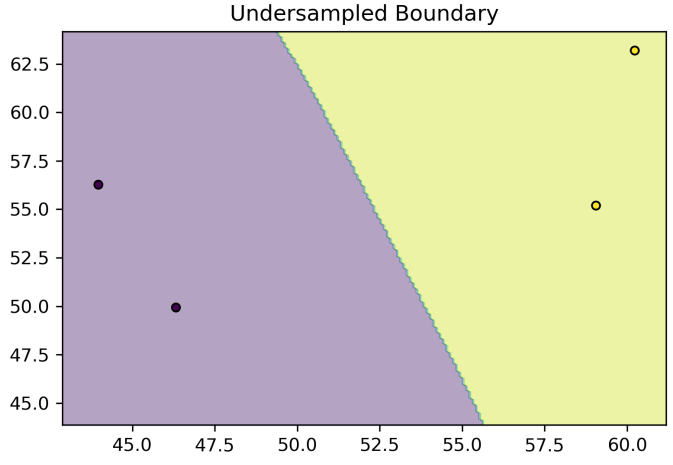






Undersampling



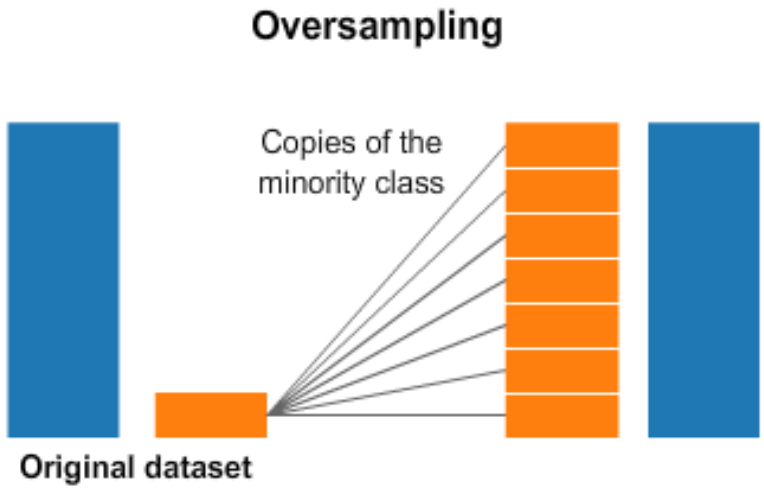


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0	1	0.5

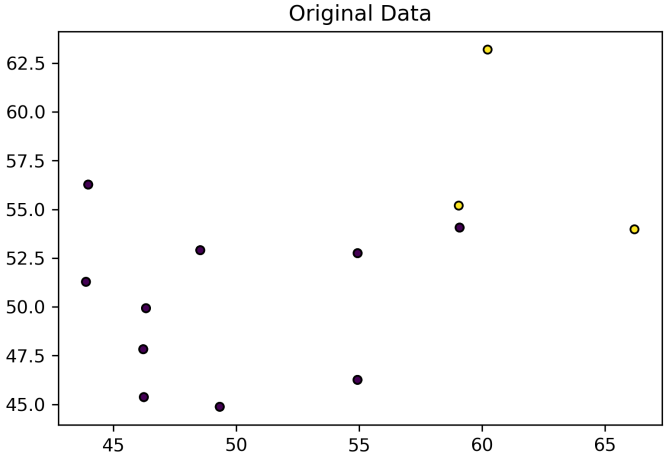
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	0.9	0.95

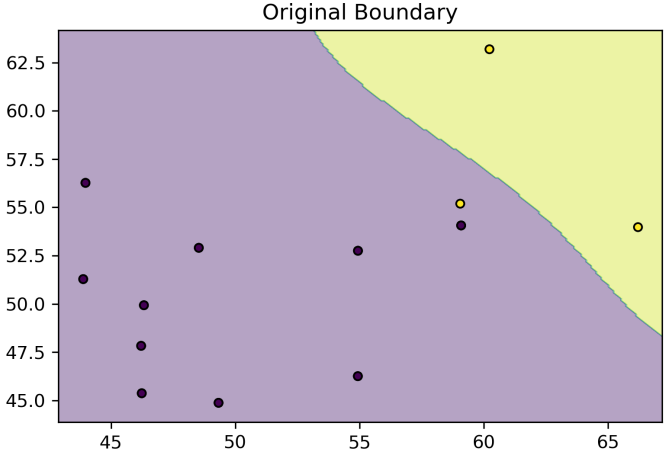
Strategy 2

- Duplicate the yellow points many times so that the data more balanced
- This is called Over-sampling

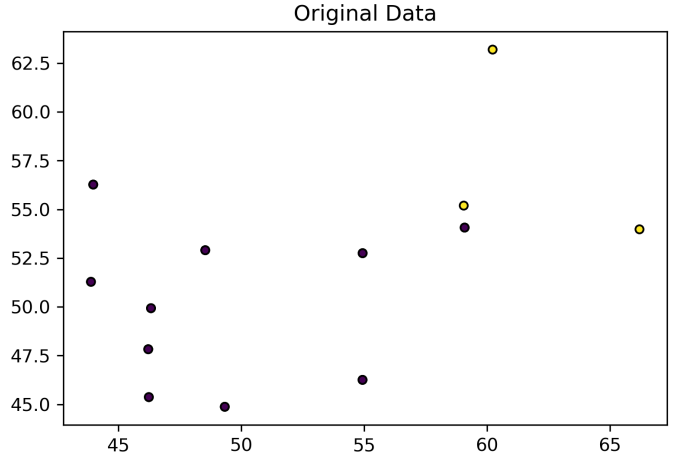


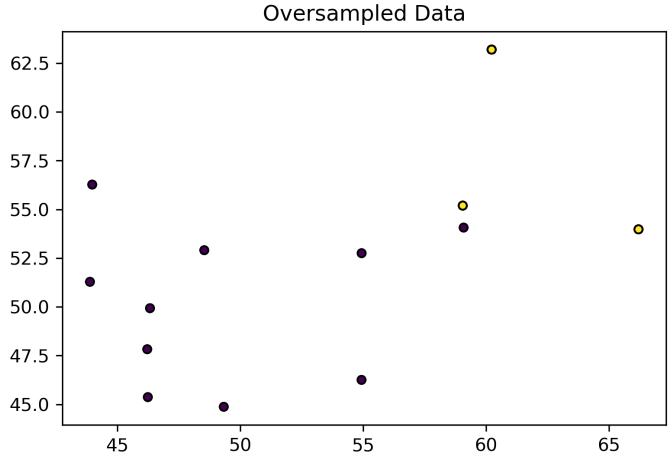
Oversampling

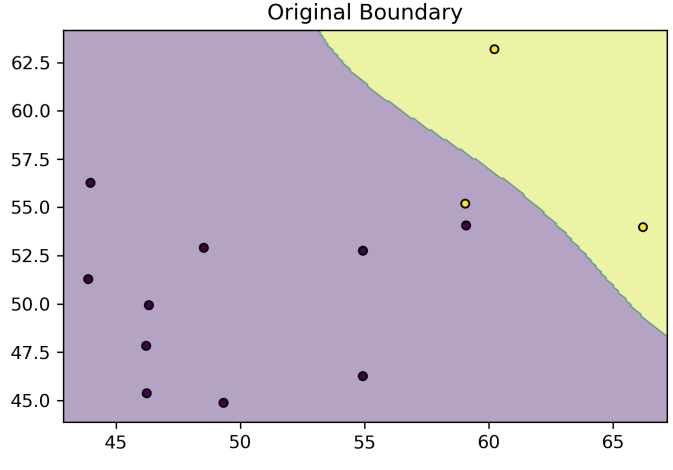




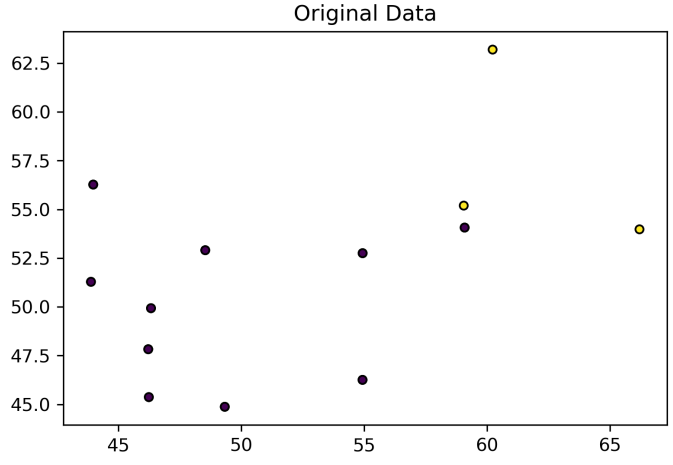
Oversampling

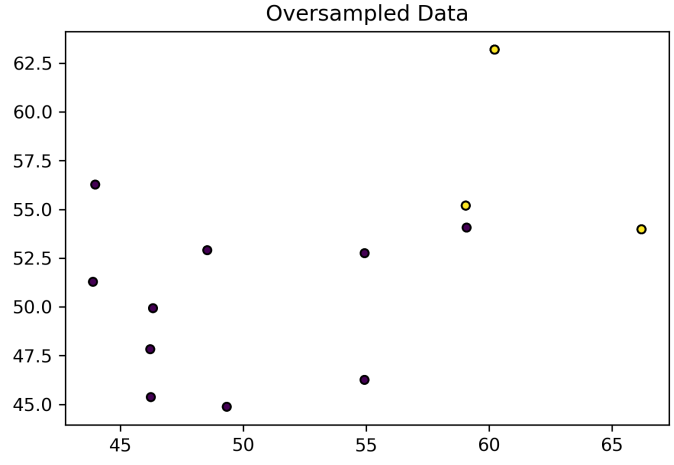


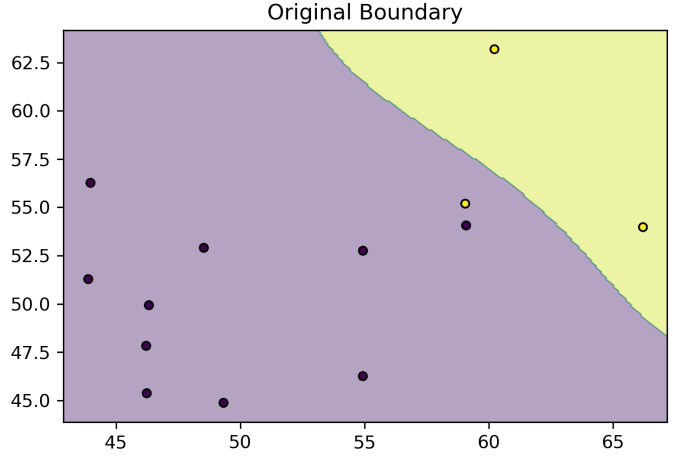


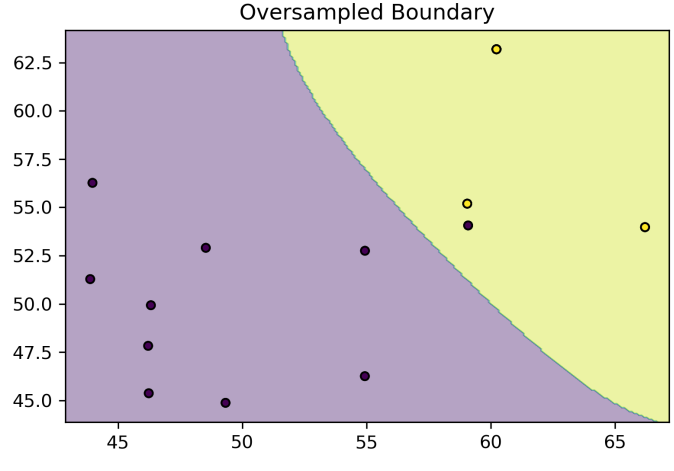


Oversampling

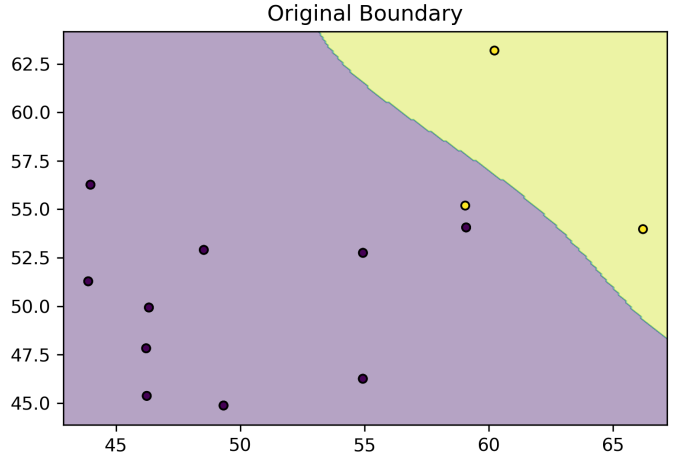


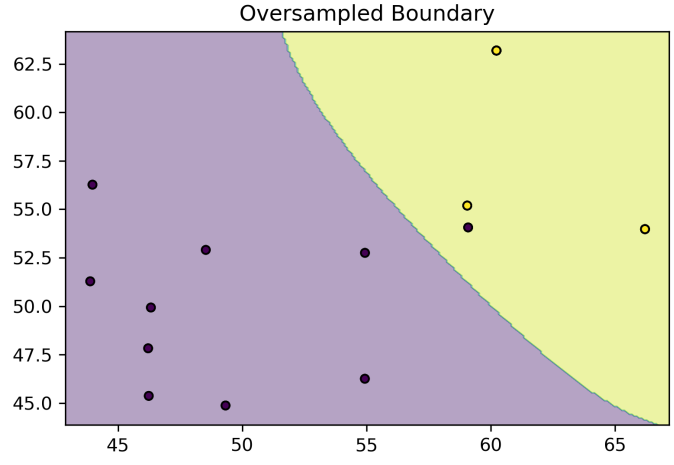






Oversampling

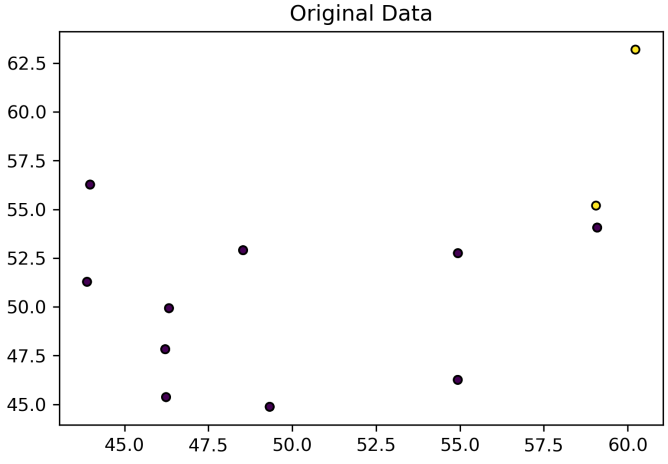


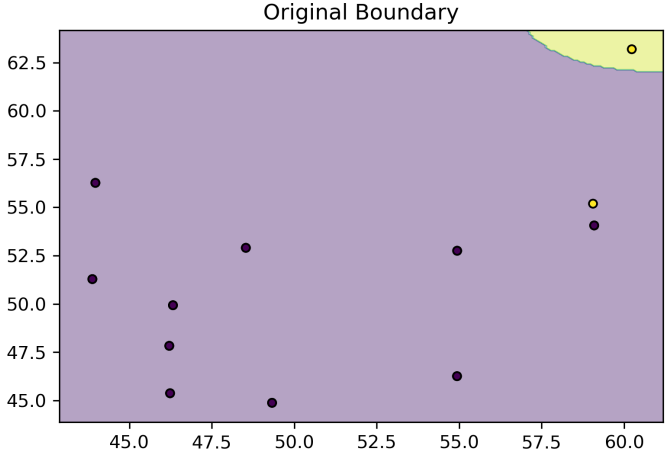


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.666667	1	0.833333

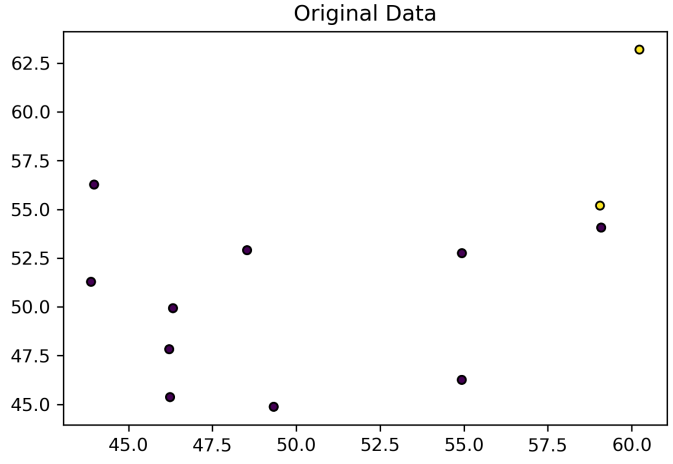
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	0.9	0.95

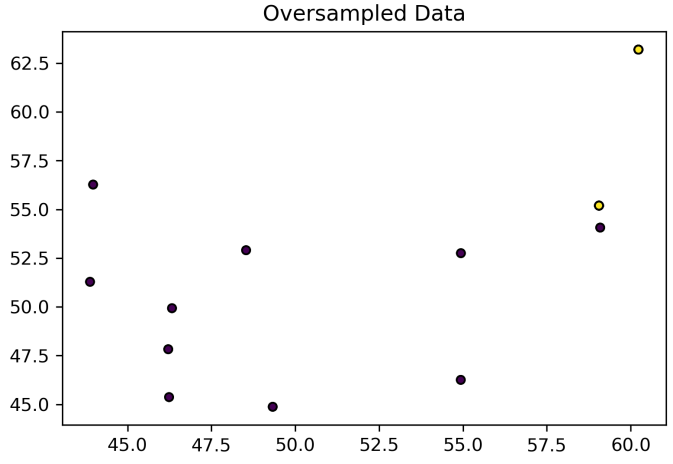
Oversampling

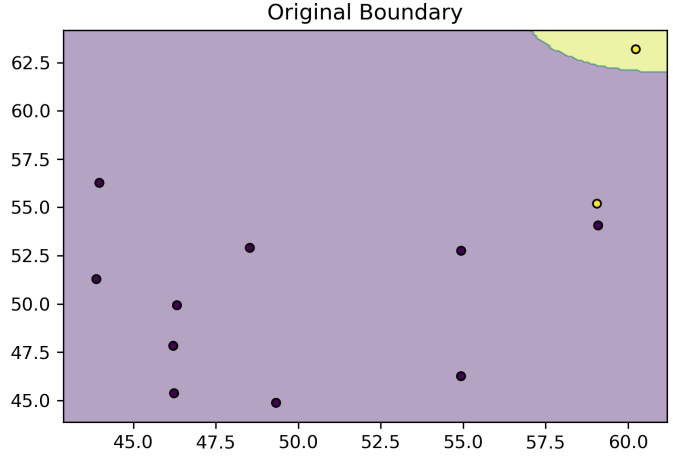




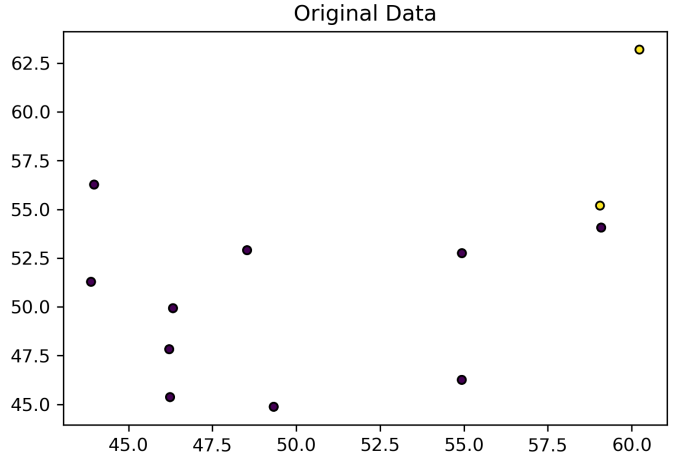
Oversampling

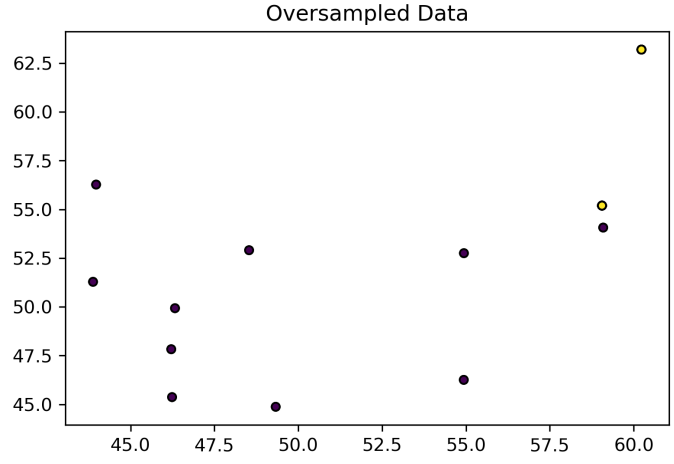


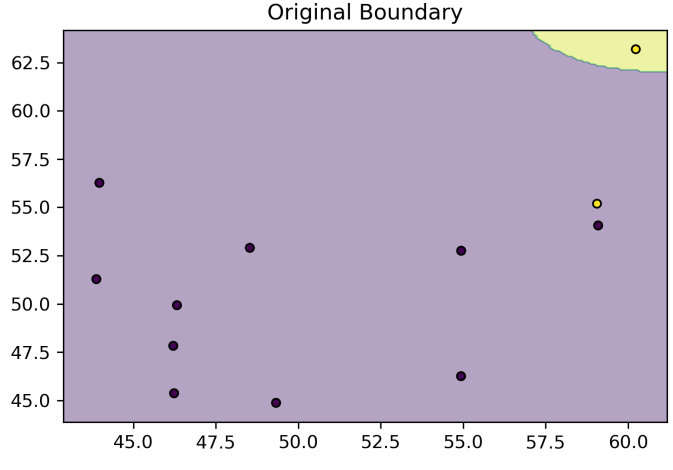


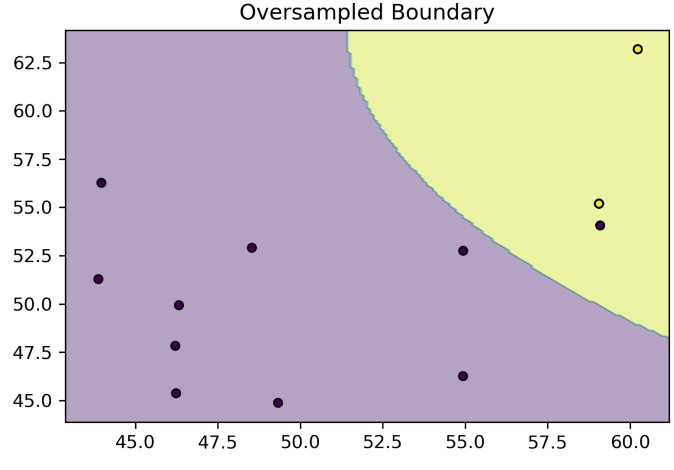


Oversampling

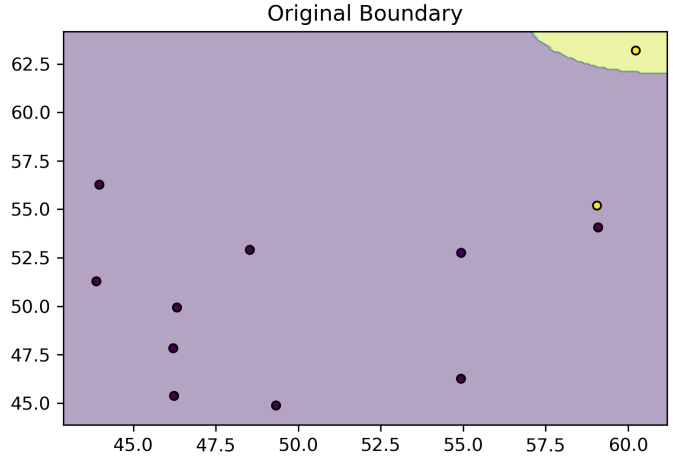


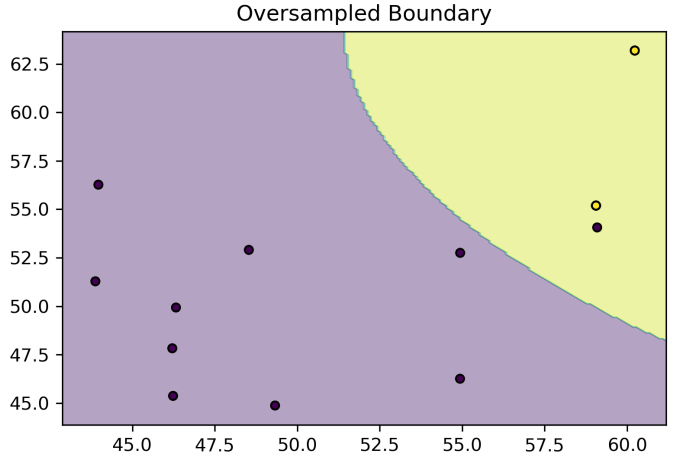






Oversampling

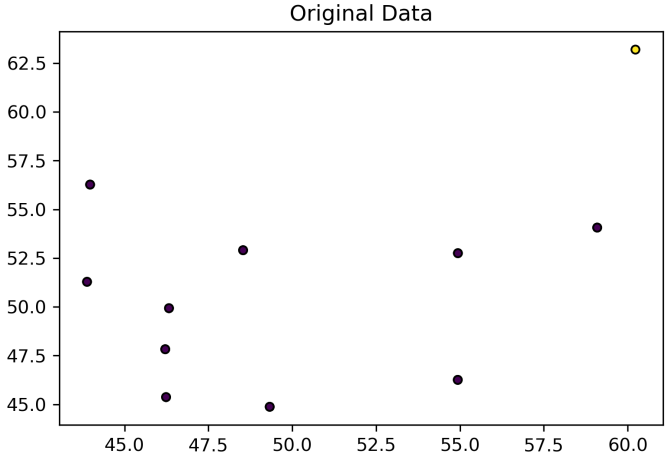


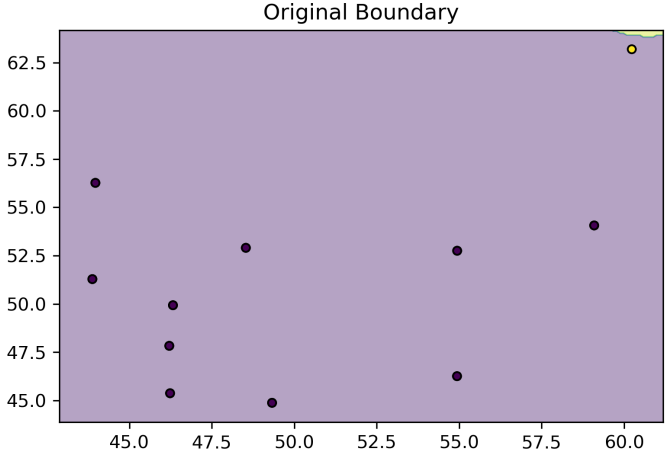


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.5	1	0.75

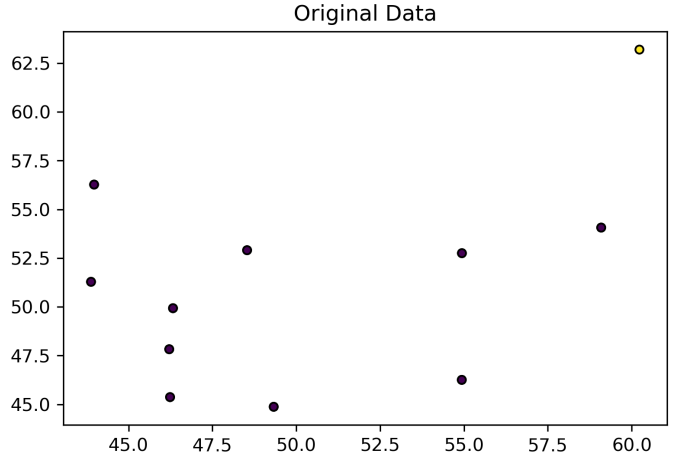
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	0.9	0.95

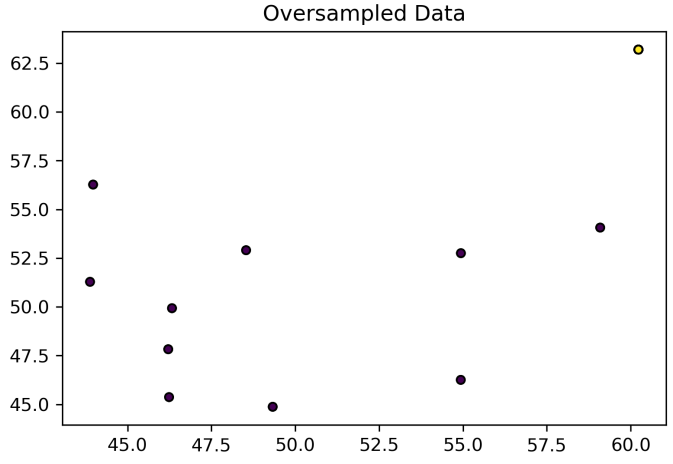
Oversampling

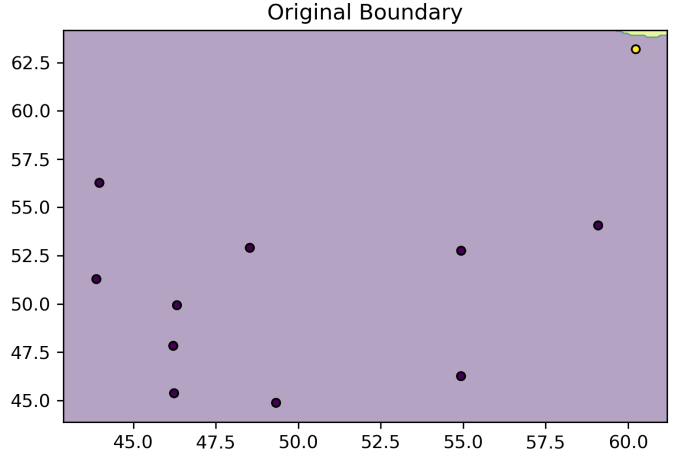




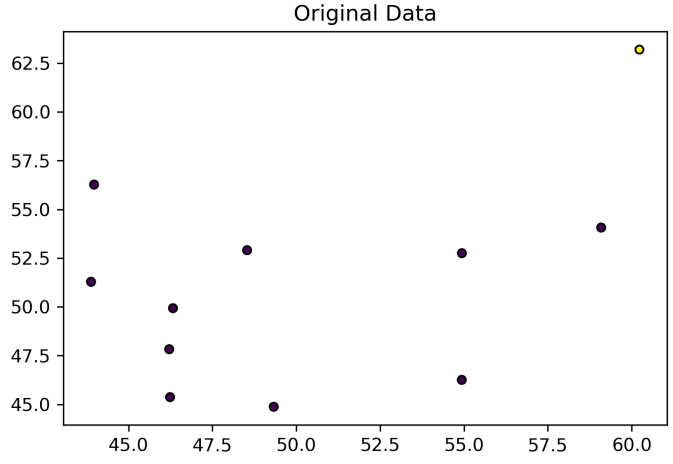
Oversampling

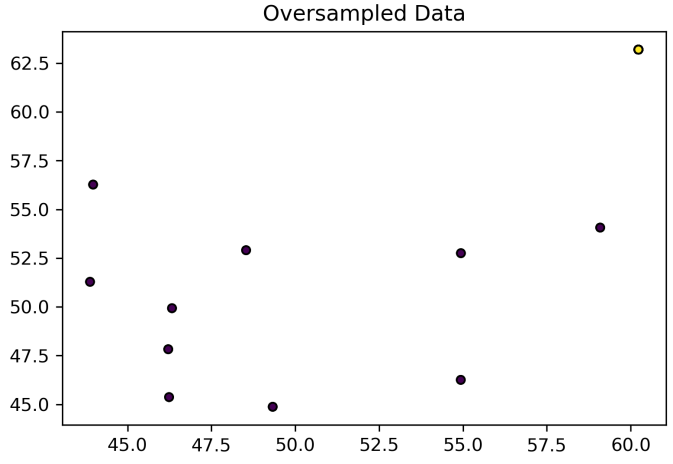


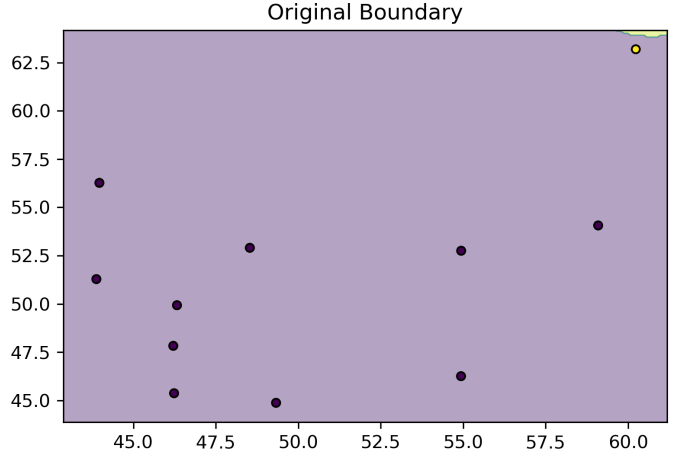


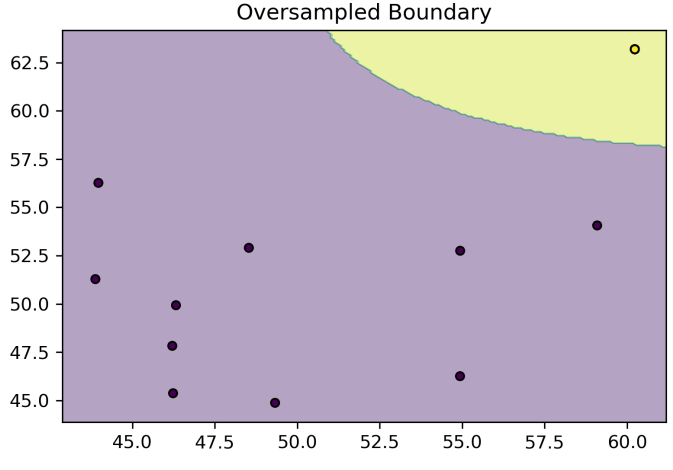


Oversampling

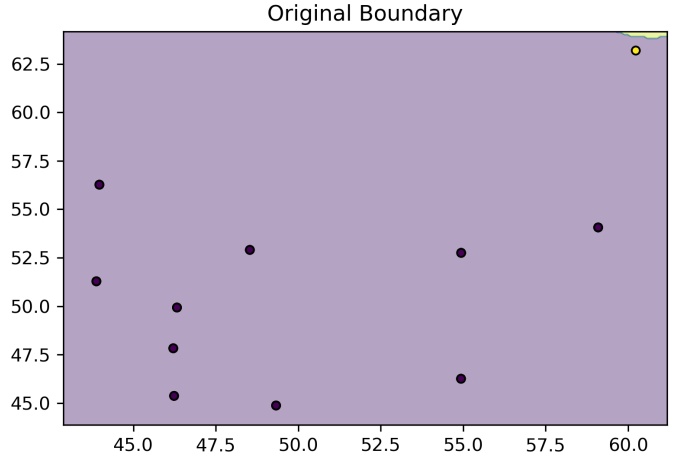


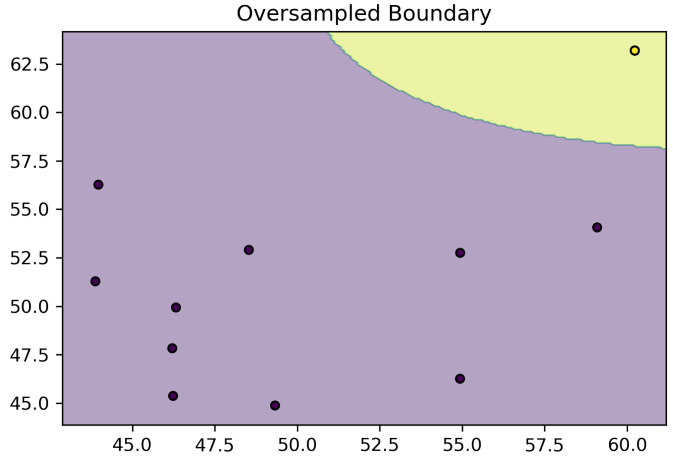






Oversampling





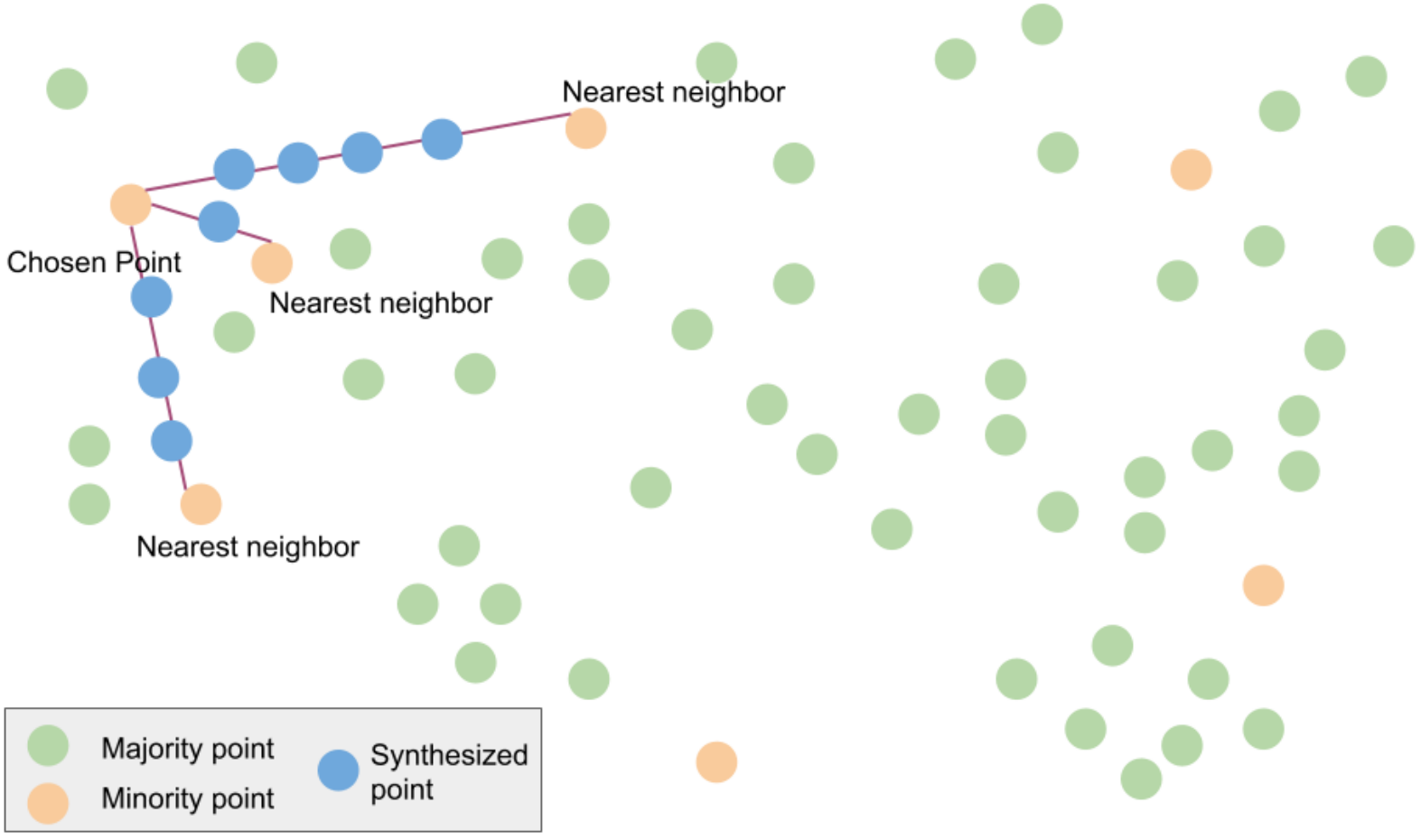
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0	1	0.5

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
1	1	1

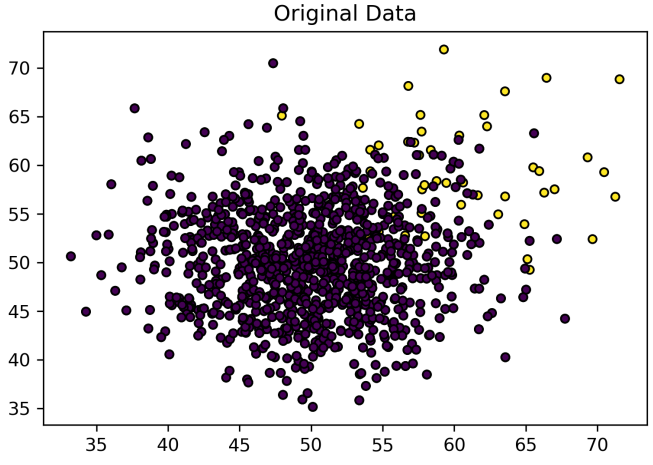
Strategy 3:

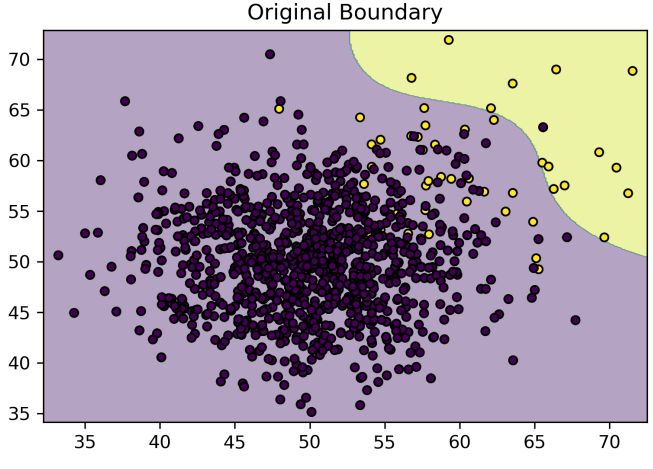
- Generate fake purple points

SMOTE

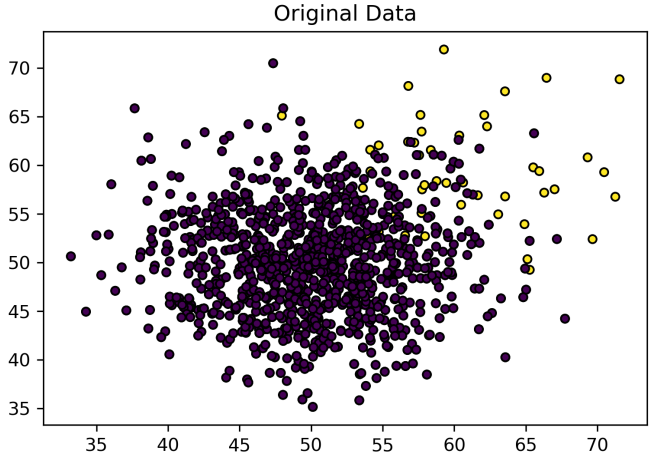

Nearest neighbor
Chosen Point
Nearest neighbor
Nearest neighbor
Majority point
Minority point
Synthesized point

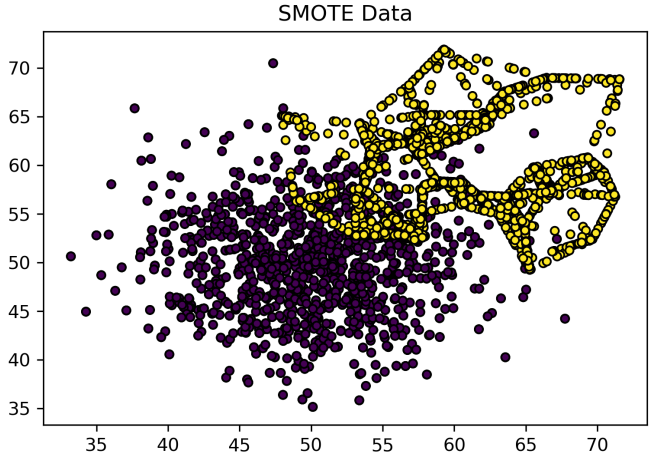
SMOTE

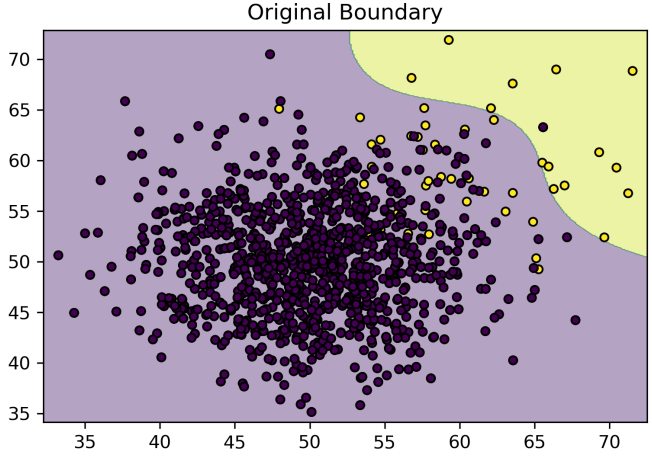

Original Data
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70


Original Boundary
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

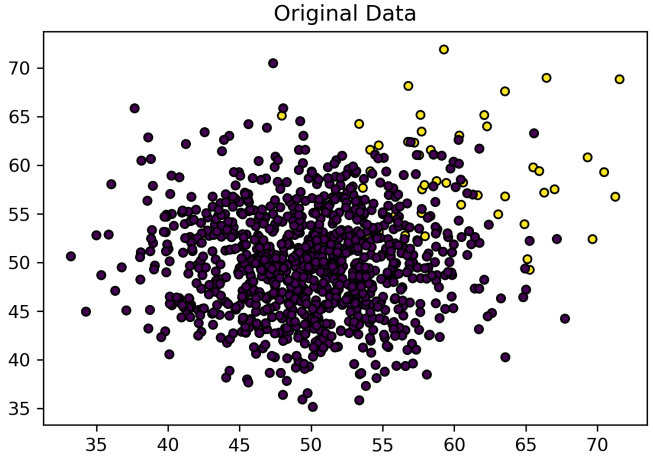
SMOTE

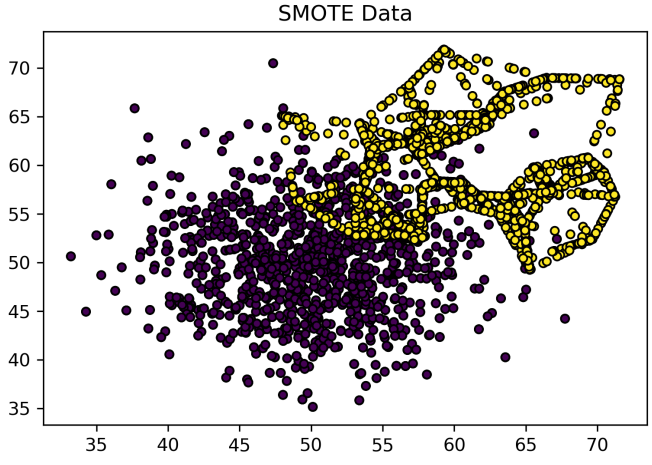

Original Data
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

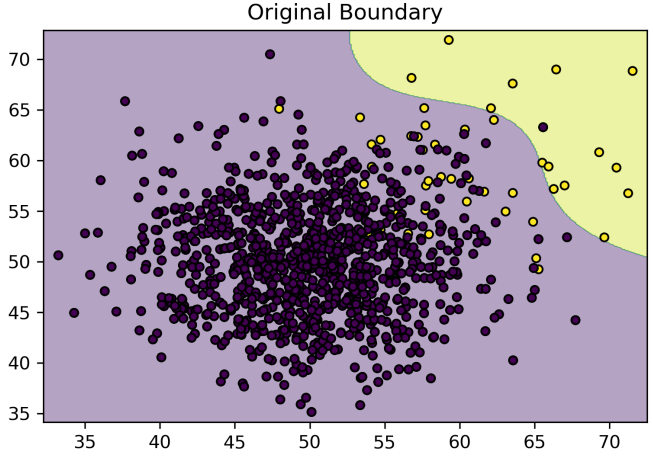

SMOTE Data
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

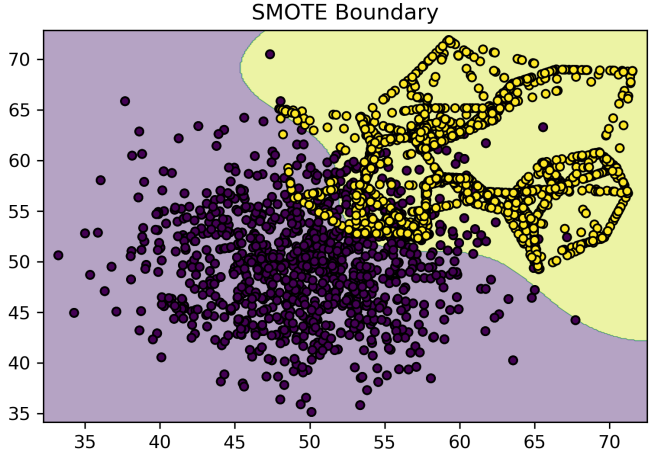

Original Boundary
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

SMOTE

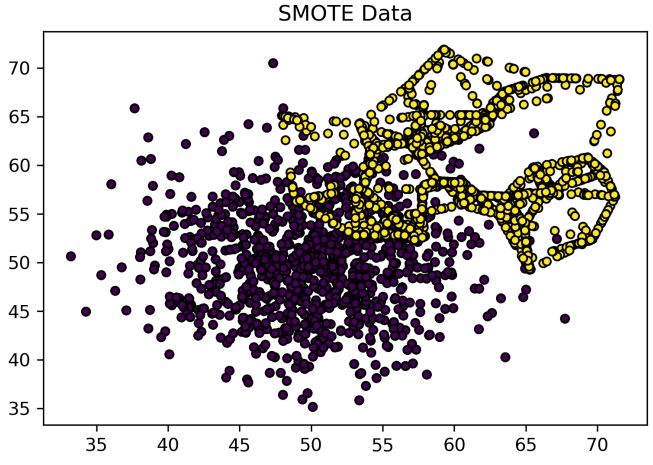

Original Data
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

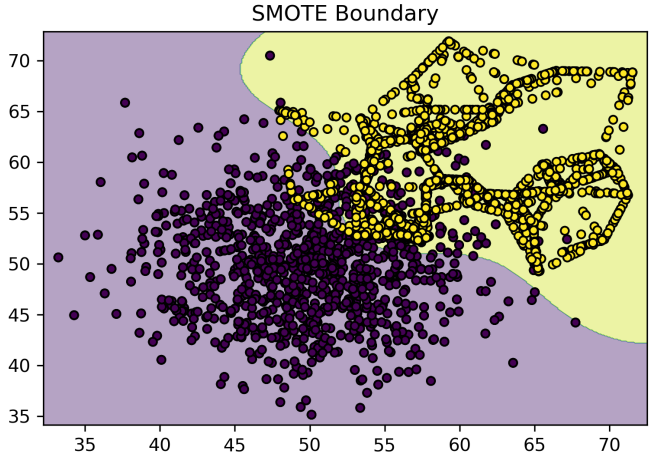

SMOTE Data
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70


Original Boundary
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70


SMOTE Boundary
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

SMOTE - Accuracy

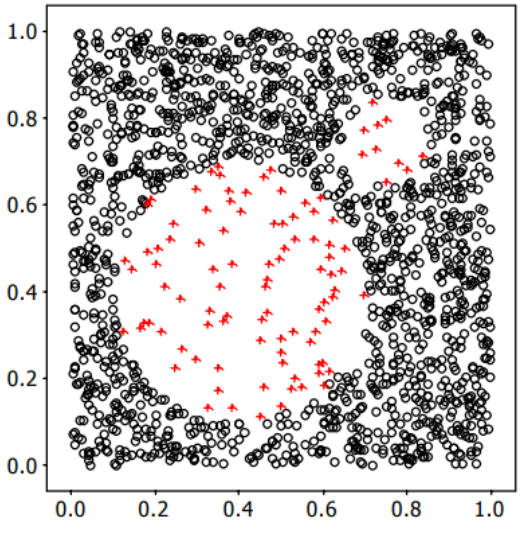




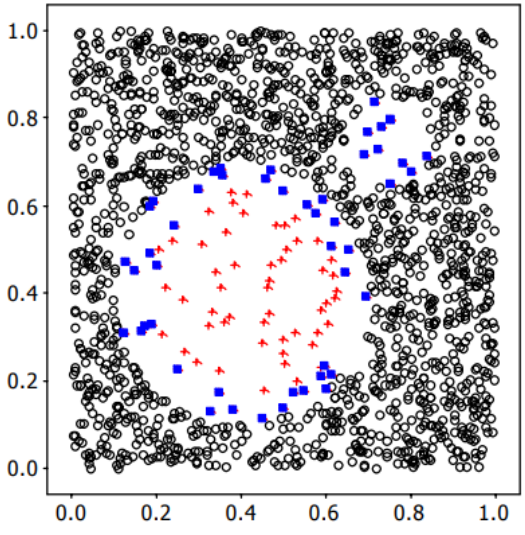
Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.28	0.999	0.6395

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.96	0.864	0.912

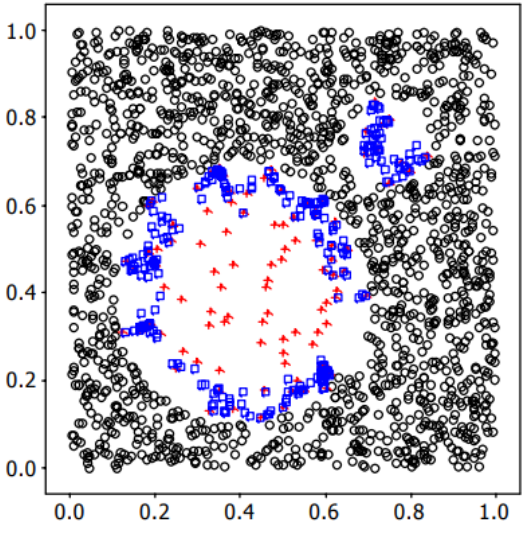
Borderline-SMOTE



The original data

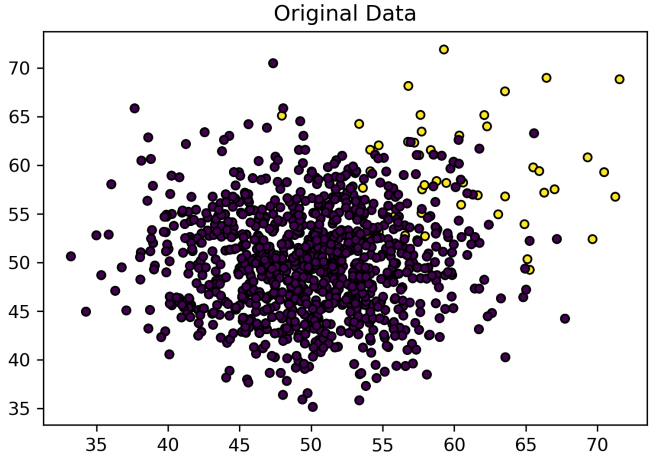


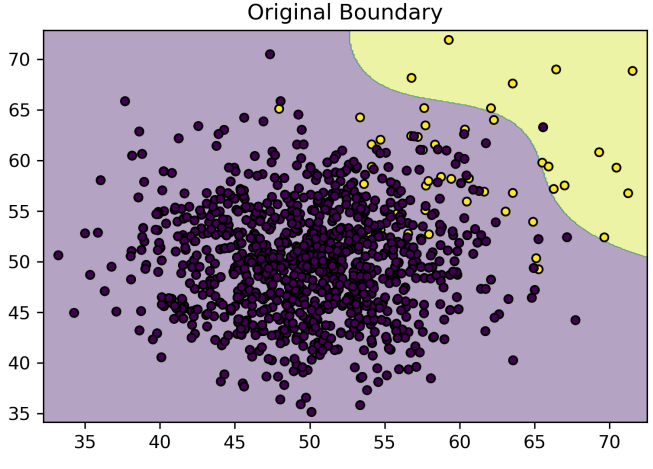
borderline minority



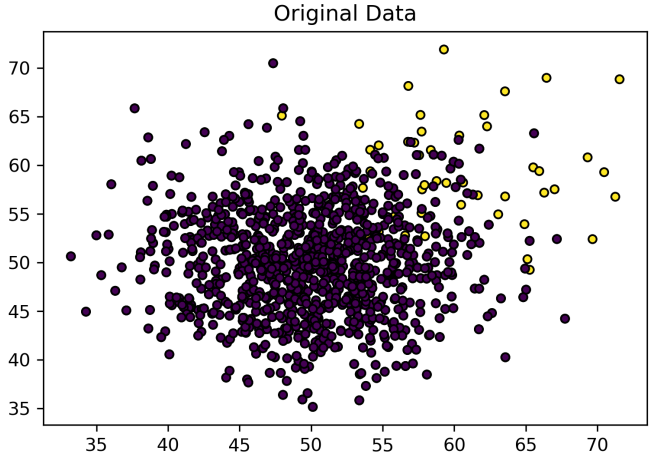
synthetic minority

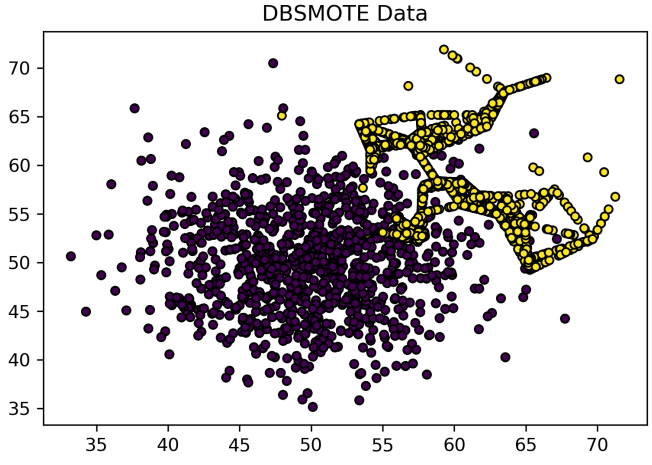
Borderline-SMOTE

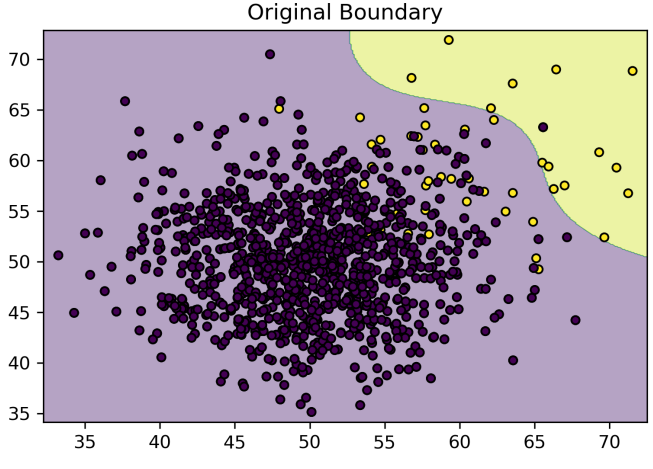




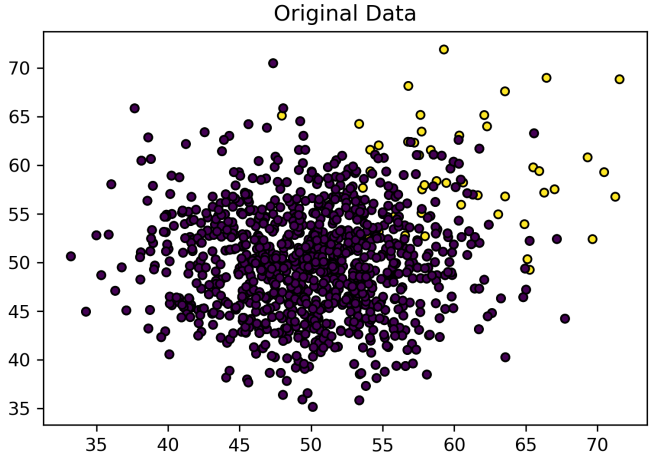
Borderline-SMOTE

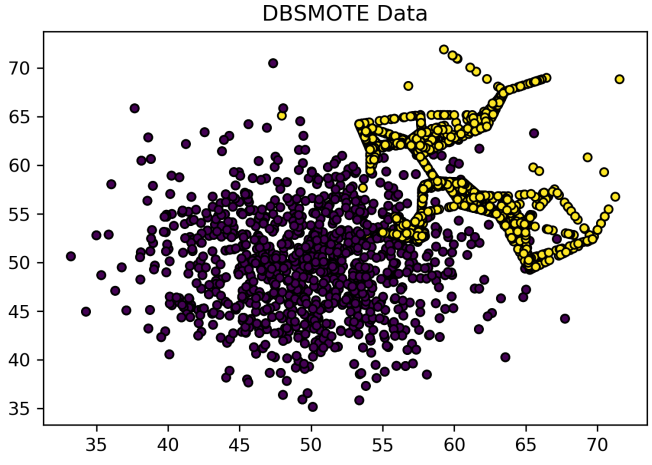


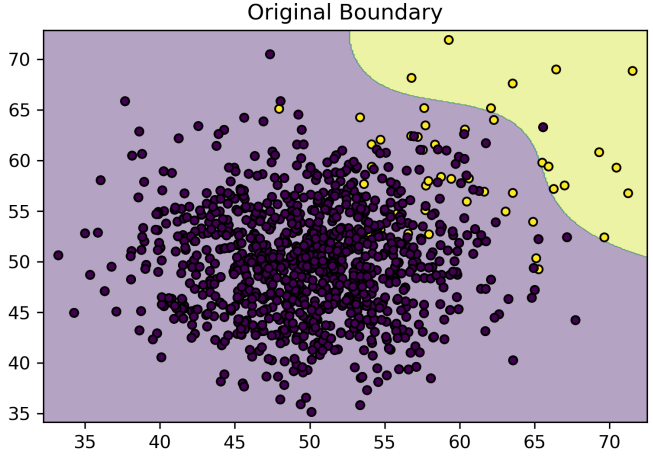


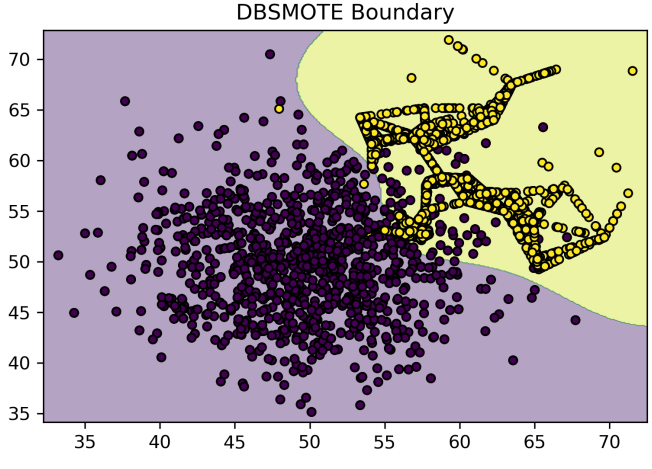


Borderline-SMOTE

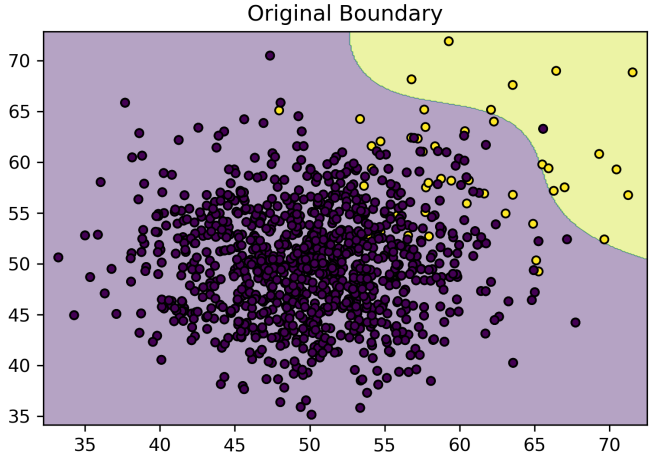


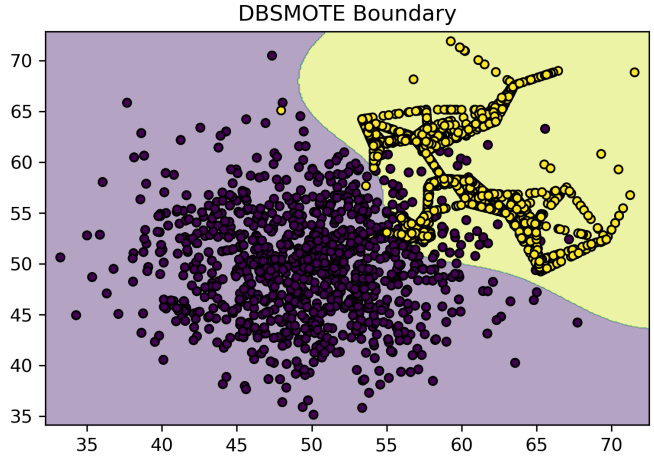






Borderline-SMOTE



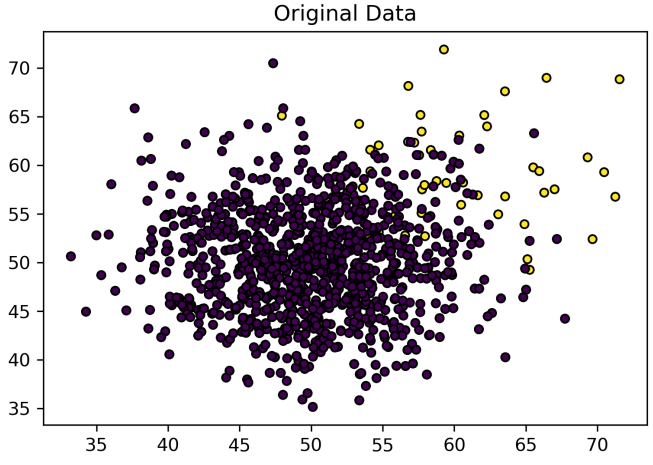


Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.28	0.999	0.6395

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.86	0.919	0.8895

Kmean SMOTE

Kmean-SMOTE


Original Data
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50
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35
40
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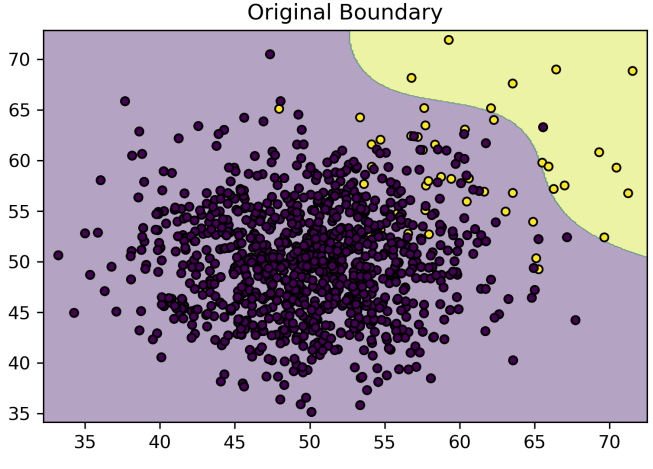
55
60
65
70


Original Boundary
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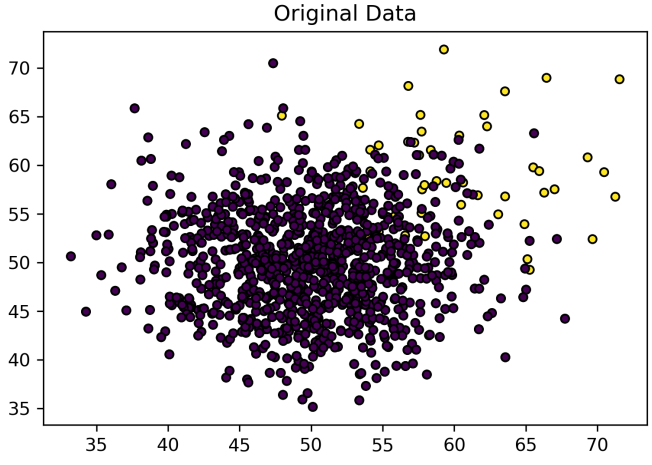
Kmean-SMOTE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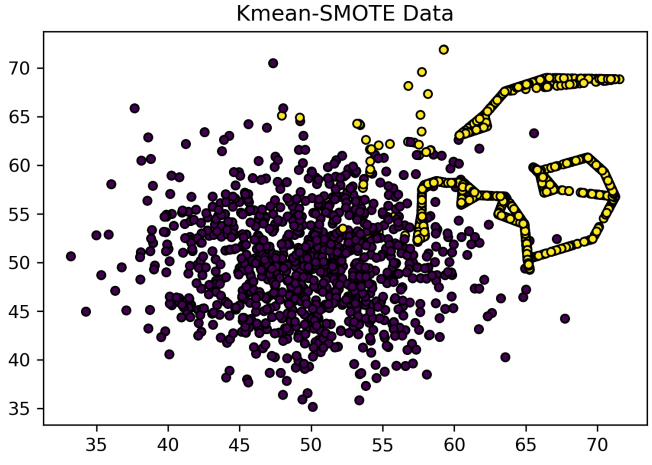
Kmean-SMOTE Data
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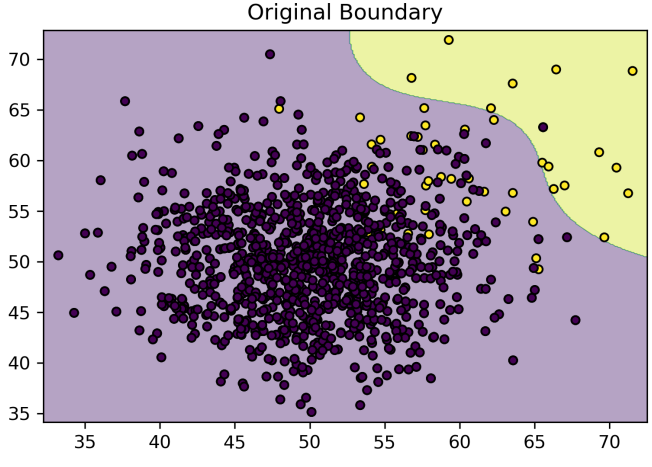
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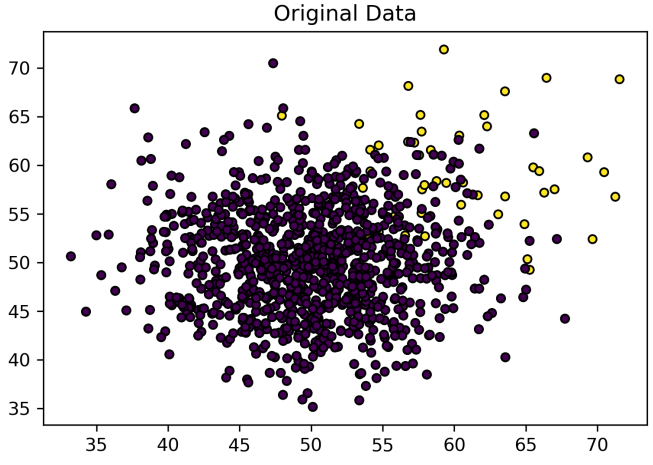
Kmean-SMOTE


Original Data
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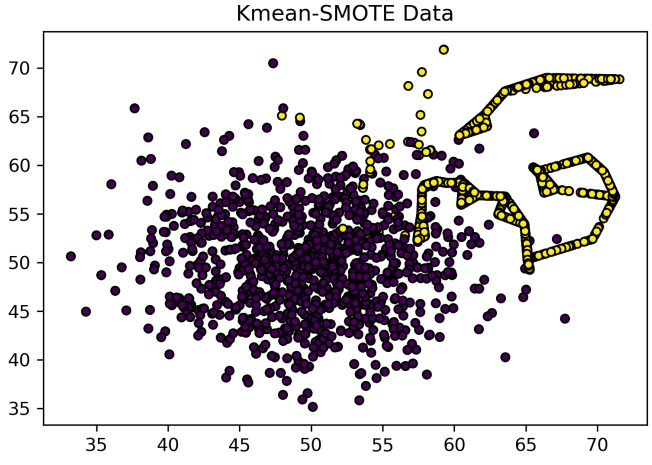
Kmean-SMOTE Data
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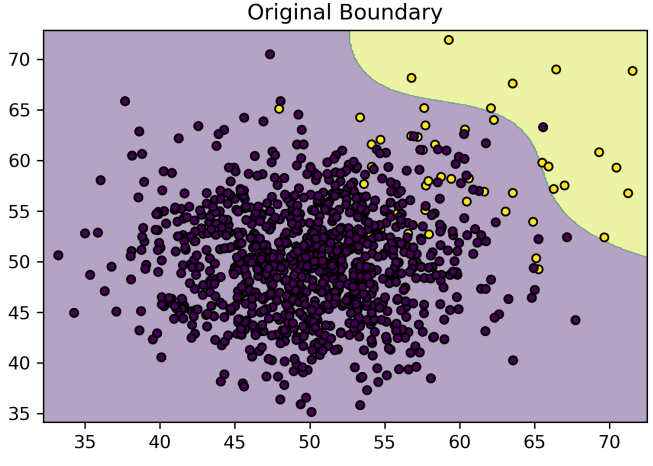
Original Boundary
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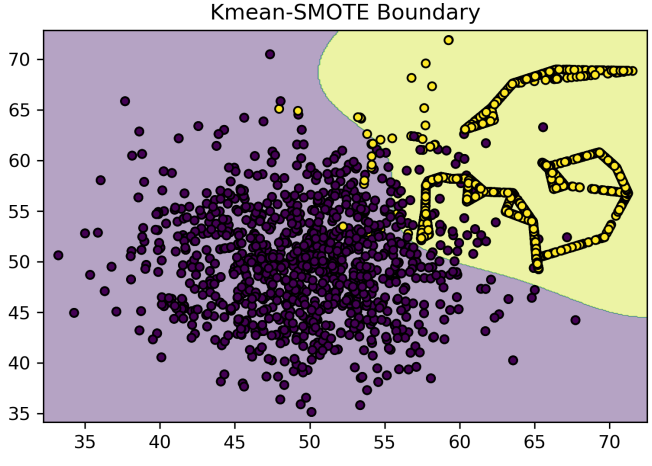
Kmean-SMOTE Boundary
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50
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40
35
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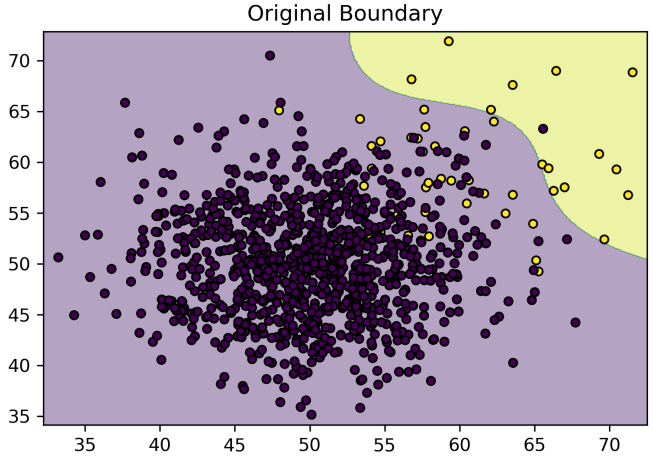
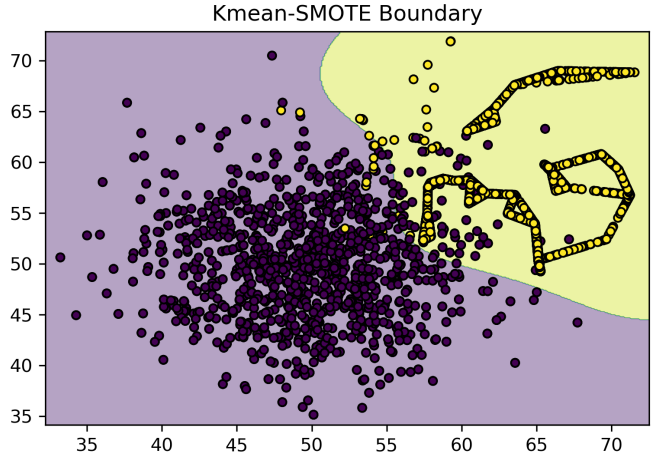
40
45
50
55
60
65
70

Kmean-SMOTE Accuracy





Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.28	0.999	0.6395

Yellow Accuracy	Purple Accuracy	Balanced Accuracy
0.82	0.94	0.88

Research Project: Data Oriented

- Find a dataset
- Propose a method
- Show that it works better than popular methods

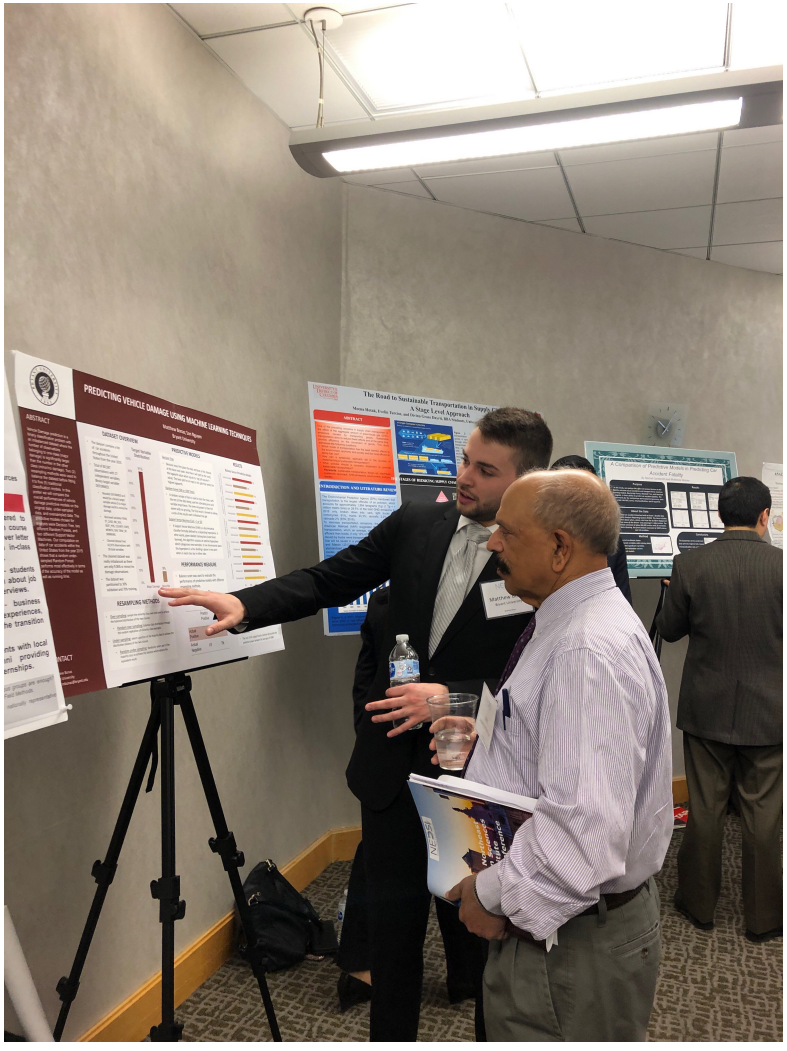
Research Project: Methodology Oriented

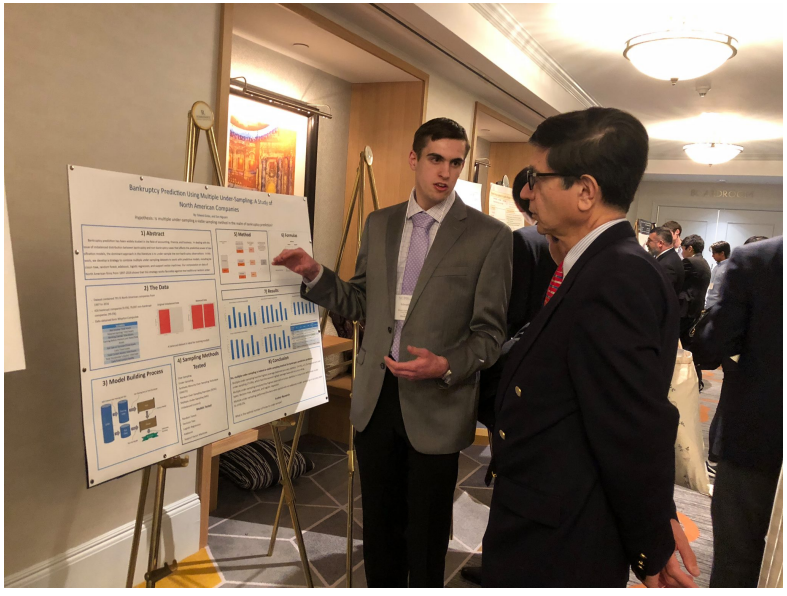
- Propose a method
- Find several datasets
- Show that your method works better than popular methods on the datasets

NEDSI 2021

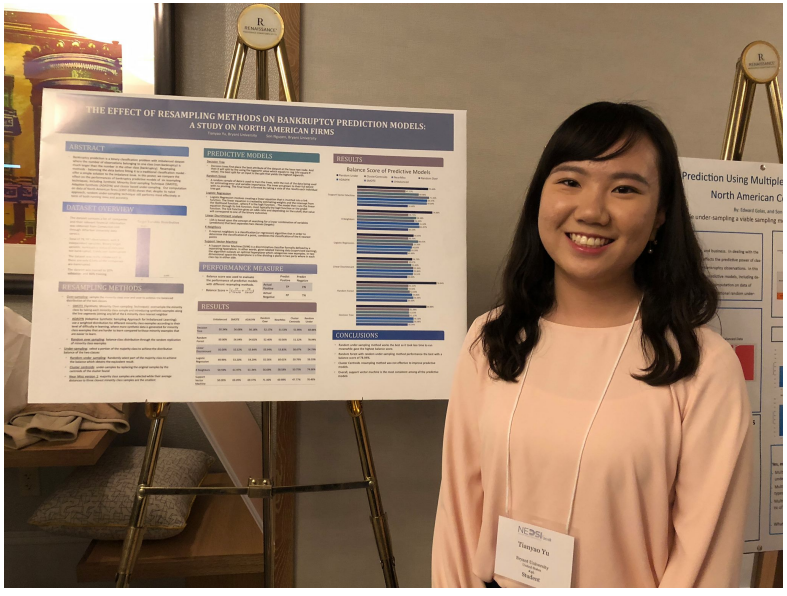
- Northeast Decision Sciences Institute Conference
- March 26-27, 2021
- Abstract Submission: January 16, 2021

NEDSI in the past





NEDSI in the past



NEDSI in the past